GAWorld v2: A Configurable Research Stack for LLM-Based Artificial Societies

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Abstract

LLM agents make it possible to study social interaction through profiles, plans, and language, yet most demonstrations stop at a single transcript or a small isolated scenario. Research on society-level questions needs persistent inhabitants whose state, memory, relationships, and exposure change across simulated time; a shared urban and institutional substrate; explicit intervention points; configurable deployments for one researcher, a classroom, or a distributed cluster; and an evidence layer that records what was run, what completed, and what did not. We present GAWorld v2, a research stack that integrates these concerns. v2 adds a 14-plugin microkernel, per-tenant world isolation, distributed execution over many machines, a parallel-worlds counterfactual engine, an Agent Playground of fourteen game-based probes, a Research Workbench for paper-to-protocol compilation, an Agent Arena for LLM-resident benchmarking, persona distillation from real people, and endogenous economics, indoor spatial reasoning, organizational governance, and audited hooks for offline citizens. The system is engineered as a configurable instrument rather than a calibrated population model: every capability is traceable to a built-in plugin, an event-bus contribution, or a documented compatibility path, and every reported number in the present paper is bound to a hashed artifact whose evidence class is explicit. We describe the architecture, audit representative capabilities against their evidence class, and state the validation, reproducibility, and ethical boundaries that govern intended use. GAWorld v2 does not validate an artificial city; it makes the construction, audit, and revision of artificial-society experiments tractable.

1 Introduction

Large language models have made it possible to construct agents that read natural-language situations, plan, and write socially legible actions. This capability has renewed interest in *generative agents* and in *LLM-based social simulation* (Park et al., 2023, 2024; Argyle et al., 2023), but a persuasive conversation is not yet an artificial society. A research-grade artificial society must let its inhabitants persist across simulated time, act from shared locations and relations, encounter consequences of earlier decisions, and leave evidence from which a researcher can reconstruct what happened. The mechanisms that surround the language model—time, space, exposure, institutions, resource constraints, audit, and intervention—must be explicit enough to inspect and vary. This is a systems problem, not a prompt-crafting problem.

GAWorld is research infrastructure for building such societies. The first arXiv system paper froze the audited K1–K5 microkernel state of the codebase at a single repository commit and reported nine built-in plugins, an artifact ledger, and a single-tenant, single-process deployment model (Project, 2026a). Since that snapshot, the system has been extended in five directions that, taken together, change what the platform is for:

1. **Plugin breadth and surface closure.** The built-in plugin set has grown from nine to fourteen; new plugins add personality-grounded behavior, dynamic interruption rules, indoor room resolution, an information-diet that mixes agent-derived queries with external sources, and organizational governance. Each plugin either participates through the event bus or is admitted as a built-in whose semantics are documented.
2. **Multi-tenant worlds.** A persistent account store with invite-code registration, role-based membership, per-world configuration overlays, copy-on-write city bundles, per-tenant quotas, and audit trails lets a single deployment host many independent societies with different visibility. When the account store is absent the system reverts, byte for byte, to its single-tenant behavior.
3. **Distributed execution and playable cities.** A world may be partitioned across a hub and many nodes: the hub owns accounts, the world metadata, and the players; each node runs its assigned inhabitants with its

own model key. Players can enter an *open* world through a browser lease, claim a resident, and act in the next tick; a strict intervention outbox routes their actions to the right node. This is a deployment primitive, not a claim of high-throughput distributed simulation.

4. **Research and counterfactual workflow.** Beyond the matched-event workflow, the stack adds a parallel-worlds engine that runs up to eight matched worlds across multiple seeds, a Research Workbench that compiles a paper or research idea into a pre-registered protocol, an Agent Arena for benchmarking residents, a policy simulation and optimization loop, and a serious-game workshop. Each result is bound to a hash, a record of completion, and an evidence class.
5. **Persona distillation and population work.** A name or a URL can be distilled into a sourced portrait that becomes a resident plus a reusable mental-model skill; offline citizens, families, organizations, and bulk imports now coexist with online ones. The City API and Agent Studio expose population construction as a first-class activity.

The platform has also internalized the lessons of its first audit. Where v1 distinguished implementation visibility from empirical validity, v2 adds explicit evidence classes (REAL, DIAGNOSTIC, SYNTHETIC, INCOMPLETE, DESIGNED, FIXTURE) and insists that every reported number carry an evidence class and a hash. Where v1 described a single notebook, v2 separates a self-contained harness (GAWorld-Bench Track A, software-path diagnostics) from a gameplay probe library (Track B, judgement-based evaluation). Where v1 described a single, frozen migration, v2 describes an open engineering frontier together with the current migration status of each subsystem.

This paper makes five contributions:

C1. An integrated, plugin-bounded research stack. We describe the v2 architecture, the boundary between the kernel and the domain-owned plugins, and the event-bus semantics that lets mechanisms observe, contribute, and filter at defined lifecycles. The architecture covers 14 built-in plugins, an agent step pipeline of 12 stages, a per-tenant world path resolver, a distributed hub-and-node protocol, and a parallel-worlds engine.

C2. Multi-tenant, multi-machine, playable societies. We show how a single deployment hosts a classroom, a research group, and a cross-machine run from the same codebase, with per-world isolation, per-tenant quotas, role-based visibility, and a hub-node message relay that routes player interventions to the correct node in at most one tick.

C3. An inspectable, evidence-classed experimental workflow. We integrate matched-event, parallel-worlds, paper-to-protocol, policy optimization, and gameplay-based evaluation under a single evidence ledger with hash-bound artifacts. Each research question has a permitted claim class, and each numerical result carries the evidence class and the producing code path.

C4. Behavior realism and societal substrate. We describe the persistent state, the Big-Five personality plugin, the dynamic-interruption module, the spatial tree that brings indoor rooms into the simulation, the endogenous inflation and credit-bound consumption model, the organizational-governance engine, and the offline-resident taxonomy.

C5. Validation, reproducibility, and ethical boundaries. We report the same five-claim validation framework as v1, expanded to accommodate the new evidence classes and the new research surfaces, and we map the engineering and research roadmaps onto the boundary between software progress and substantive social-scientific claim.

GAWorld v2 is presented as a configurable scientific instrument, not as a representative synthetic population, a digital twin, or a forecasting system. Its artifacts do not establish that generated agents reproduce particular people, that aggregate trajectories match a real city, or that a simulated intervention estimates a real policy effect. We instead treat the system as an instrument whose outputs are evaluated at five claim levels (individual consistency, macro correspondence, emergent dynamics, counterfactual response, reproducibility) and whose evidence ledger is a prerequisite for substantive interpretation.

The remainder of the paper proceeds as follows. Section 2 states the design goals; Section 3 positions v2 against generative agents, language agents, social evaluation, agent-based modeling, and platform research. Sections 4 to 6 describe the agent, the kernel, and the cognitive substrate. Sections 7 to 9 describe the societal substrate, multi-tenancy, and the distributed playable runtime. Sections 10 and 11 describe the research and gameplay surfaces. Sections 12 and 13 audit the existing capability cases against the five claim levels. Section 14 states the limits and the roadmap. Section 15 concludes.

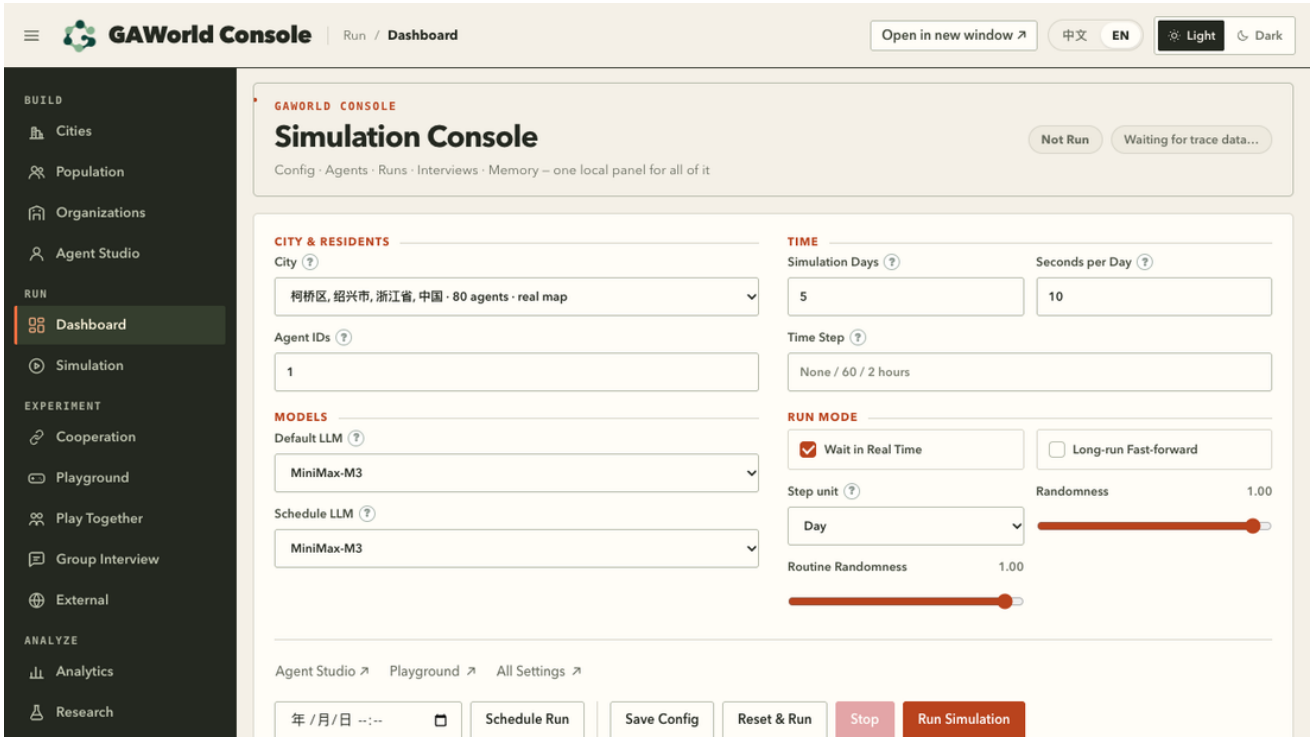


Figure 1: The GAWorld Console. The left navigation exposes every v2 surface: *Cities* and *Population* build the city bundle; *Organizations* edits rules and members; *Agent Studio* shapes individual residents; the *Dashboard* (shown) configures and starts a run; the *Playground*, *Play Together*, and *Group Interview* tabs run game-based probes, an open-world player lease, and one-shot elicitation; *External* drives the relay and external-environment services; and *Analytics* / *Research* surface the result inspection and the counterfactual / paper-to-protocol workflows. The screenshot was captured against the bundled Keqiao (Shaoxing) city bundle in single-tenant, single-process mode; the dashboard UI was switched to English via `window.setLocale('en')`, while the resident content reflects the language of the bundled data.

2 Design Goals

GAWorld v2 is intended as a configurable research instrument rather than as a claim that a language model is, by itself, a valid population model. Its design goals therefore concern properties that a researcher can inspect in the running system, and each property has a concrete mechanism in the present implementation and a boundary that prevents the mechanism from being read as an empirical guarantee. Where v1 listed six properties, v2 lists eight, adding two that the v1 platform could not satisfy: *per-tenant world isolation* and *human participation under the same first-class intervention surface as scheduled events.*

2.1 Persistence across simulated time

An agent must carry consequences from one decision context into later ones, rather than being reconstructed from a stateless prompt at every tick. v2 combines a mutable state record with file-backed episodic memory, daily logs and diaries, schedules, and longer-lived relationship and growth records. A stateful run restores the day counter and per-agent memories; the memory lifecycle can retrieve, consolidate, summarize, and decay stored episodes. Persistence is therefore testable: a researcher can inspect both the state before and after a tick and the files used in later recall. Persistence does not establish psychological fidelity. Many memories and summaries are language-model outputs, configuration can disable lifecycle operations, and several archived cross-phase evaluations are incomplete.

2.2 Situatedness in a shared world

Actions should be conditioned on where and when an agent acts and on what is locally observable. The simulation clock exposes day, time, and tick index; the city map represents named locations, connections, and indoor rooms; and environment, local-physical, social, policy, and information mechanisms contribute context to the cognition loop. Movement updates location before later stages use the new setting, social influence is computed over the current network rather than over an unconstrained cast of characters, and indoor rooms are resolved against the same activity the agent is performing. v2's addition of an *away* state (Project, 2026m) lets inhabitants temporarily leave the city and return to a specific home node. This is situated computation, not a calibrated digital twin: map detail, physical dynamics, agent access to information, and the indoor-room tree are abstractions selected by configuration.

2.3 Composability without concealing control flow

Researchers should be able to change an experimental mechanism without rewriting the complete agent loop. v2 provides three bounded extension surfaces: a configurable stage sequence for cognition, an event bus with observe/contribute/filter semantics, and a registry for lifecycle-managed plugins. The audited built-in set now contains fourteen plugins: intervention, skills, interests, life events, economy, local physical perception, real work, dynamic behavior, spatial preferences, personality, indoor rooms and venue capacity, organizations, governance, and offline citizens. These join existing memory, environment, and social-network subsystems that retain direct compatibility paths. Each plugin declares its dependencies, owns configuration namespaces and per-agent state, and contributes through the bus or through lifecycle hooks. Composability is not unrestricted hot swapping: the stage pipeline exchanges a shared step dictionary, and three legacy subsystems (memory, environment, social network) retain direct paths. The audit table in section 5 lists the boundary state of each subsystem.

2.4 Controllability for counterfactual experiments

A research run should expose explicit points at which a scenario or intervention is introduced. Configuration fixes profiles, city assets, random seed, mechanism flags, schedules, and policy events. The intervention plugin contributes a recommendation feed through the event bus and records its domain-specific metrics. The Controller API validates structured movement requests through a priority-ordered validator chain and exposes audited runtime interventions for state mutation, configuration update, and agent removal. v2 adds three new controllability surfaces: a parallel-worlds engine that runs up to eight matched worlds with a shared seed; a Research Workbench that compiles a research idea or a paper into a pre-registered protocol with pre-specified outcomes and minimum-effect thresholds; and a policy-simulation loop that proposes a candidate policy, simulates, scores, and iterates. None of these are guaranteed to identify a real-world effect; they are designed to make the construct, the contrast, and the completion status of every arm visible to the analyst.

2.5 Inspectability from decision to artifact

A claim about an artificial society should be traceable to artifacts below the level of a prose summary. v2 writes per-agent logs and memories, environment timelines, state histories, plugin-owned outputs, and experiment comparisons; its interfaces expose portions of those records for inspection and replay. The kernel Recorder additionally implements a clock-stamped JSONL stream and records Controller interventions and denials. v2 adds two new inspectability surfaces: a parallel-worlds results file that records per-arm completion status, per-resident ATE, bootstrap confidence intervals, and sign-flip p-values; and a per-tenant evidence ledger that binds every reported number to a hash and an evidence class. The unified Recorder is not yet the main action/trace path: the established simulator continues to write several specialized formats directly, and the harness-built benchmark is a software-path diagnostic, not a replayable trace.

2.6 Backward-compatible modularization

The system must remain usable while a monolithic research prototype is decomposed. A typed `Agent` view shares, rather than copies, the legacy dictionary; `EventBus` preserves the legacy hook signature; and the 12-stage pipeline was carved from the existing per-agent step while retaining the shared step dictionary. v2 adds two new compatibilities: the `AccountStore` is optional (its absence reverts the system to single mode), and the `DistributedRelayBackend` is optional (its absence falls back to the in-process relay). These adapters allow

migrated and legacy modules to coexist and make migration state observable. They also retain weakly typed surfaces and shared mutable state. Backward compatibility is a transition strategy, not evidence that modularization is complete or that every old configuration produces an identical trajectory.

2.7 Per-tenant world isolation

A single deployment should host many independent societies, each with its own city bundle, its own configuration overlay, its own run, its own output tree, and its own visibility. v2 implements this by treating a *World* as a first-class object: a city bundle plus a per-world configuration overlay plus a per-world output root. The active world is resolved from a cookie, validated against the caller's role, and propagated through a context variable. All path-resolving functions (`effective_config`, `state_csv_path`, `records_dir`, ...) consult the active world. When the account store is absent, the single `default` world is the only world and the system behaves as it did before multi-tenancy was added. A change in a world's output path is therefore visible at the request level rather than at a process-restart level. Per-tenant isolation does not establish that two tenants running identical configurations will produce identical trajectories; the underlying model, the random seed, and the cross-process timing of distributed runs remain sources of variation.

2.8 Human participation as a first-class intervention

Researchers and the public should be able to enter a society, claim a resident, and act under the same first-class intervention surface that scheduled events use. v2 implements this through a *player lease*: browsers request a two-minute lease, see current perception, send a bounded action, and release the resident. In a multi-tenant deployment, the lease is routed to the node that owns the resident; in a single-process deployment, the lease is local. The intervention type is the same, the audit trail is the same, and the next-tick perception includes the player's action. This makes human participation inspectable in the same ledger as scheduled events, but it does not validate that the human's action faithfully represents the claimed resident. The same evidence-class rules apply: a player's claim is a real, inspectable intervention, not a ground truth.

3 Related Work

GAWorld v2 draws on five bodies of work that answer different parts of the artificial-society problem. Its contribution is a system integration and an audited set of research surfaces; we do not claim that the underlying ideas of memory, tool-using agents, social evaluation, agent-based validation, or agent platforms are unique to GAWorld.

3.1 Generative agents and simulated individuals

Generative Agents demonstrated that an LLM-driven architecture combining an experience stream, retrieval, reflection, and planning can sustain socially legible behavior in a sandbox community (Park et al., 2023). Later work grounds agents in extensive self-report data and evaluates how well responses predict held-out answers from the same individuals (Park et al., 2024). Research on "silicon samples" explores conditioning language models to approximate distributions of survey responses (Argyle et al., 2023). These studies motivate profile grounding and longitudinal state, but they also distinguish two validation targets: internally coherent simulated inhabitants and externally evaluated models of particular people or populations.

GAWorld v2 is oriented toward constructing the surrounding research system. It adds configurable urban, social, economic, work, organizational, information, and policy mechanisms; multiple operational interfaces including a class-scale multi-tenant deployment; and explicit experiment and artifact workflows around a persistent-agent loop. Its archived profiles and traces have not undergone the held-out, person-level validation used in self-report-grounded agents. Accordingly, v2 adopts persistence as a system property and treats fidelity as an open empirical claim, as the v1 system paper already argued (Project, 2026a).

3.2 Language-agent cognition and architecture

LLM-agent research has developed reusable patterns for coupling language generation to action. ReAct interleaves reasoning and environment-facing actions (Yao et al., 2023); Reflexion stores verbal feedback to improve later attempts (Shinn et al., 2023); Voyager combines iterative prompting, environment feedback, and an accumulating

skill library in an open-ended embodied setting (Wang et al., 2023). MemoryBank studies long-term memory with retrieval and forgetting mechanisms (Zhong et al., 2024), while CoALA offers a conceptual organization of language agents in terms of memory, action space, and decision procedures (Sumers et al., 2024). CAMEL examines role-conditioned communication among language agents (Li et al., 2023).

v2 informs its separation of structured state, episodic storage, retrieval, planning, action, reflection, and learned skills from these systems. It differs in scope rather than by proposing a new generic reasoning algorithm: it connects such components to a persistent multi-agent society, a clocked file-backed runtime, an event bus, a plugin registry, a per-tenant world model, and a parallel-worlds engine. v2 additionally adopts a *pluggable personality* layer so the cognitive profile of a resident can be replaced without rewriting the rest of the loop. The contribution is an implemented integration for social simulation, not a claim of a universally superior cognitive architecture.

3.3 Social-agent tasks and evaluation

AgentBench evaluates LLM agents across interactive environments and emphasizes the gap between language ability and reliable action (Liu et al., 2024). AgentBoard adds fine-grained progress measures and analytical visualization for multi-turn agents (Ma et al., 2024). SOTOPIA focuses specifically on social intelligence by placing agents in interactive social scenarios and evaluating goal achievement and social dimensions (Zhou et al., 2024). More recent work proposes social benchmarks that go beyond pairwise interaction, including GroupBench for collaborative social reasoning and the social-cognition probes discussed in Hashimoto et al. (2022) and Leng and Yuan (2024).

v2 addresses a complementary unit of analysis: a persistent society run whose agents, environment, institutions, and artifacts evolve across simulated time. Its benchmark layer can check selected outputs, known-sign diagnostics, placebos, and determinism when suitable artifacts exist. v2 adds two new evaluation surfaces: the *Agent Playground* of fourteen game-based probes, in which a small group of residents faces a structured scenario under deterministic scoring rules; and the *Agent Arena*, in which a fixed set of residents answers a controlled question bank under correctness and latency metrics. Neither is a replacement for standardized social-intelligence benchmarks; they are probes that complement population-scale traces.

3.4 Agent-based social simulation and validation

Agent-based modeling (ABM) treats macro patterns as outcomes of specified micro-level rules and interactions, while generative social science emphasizes using such models to explain how collective regularities can arise (Epstein, 2006). The ODD protocol provides a standard vocabulary for documenting an individual- or agent-based model's overview, design concepts, and details (Grimm et al., 2006). Methodological work on empirical ABM validation stresses that matching selected aggregate observations is not sufficient: calibration, identification, sensitivity, multiple validation targets, and out-of-sample evidence remain distinct concerns (Windrum et al., 2007; Fagiolo et al., 2007). The recent *Human-AI interaction* literature emphasizes the gap between model behavior and population behavior when the same model is reused in different contexts (Kapania et al., 2025).

v2 inherits this demand for explicit mechanisms and layered evidence, but LLM mediation introduces additional sources of variation. Prompt context, provider behavior, model versions, parsing fallbacks, and generated text can change trajectories even when conventional parameters and seeds are recorded. v2 therefore couples ABM-style configuration and intervention design to provider logs, raw agent traces, per-arm completion status, an artifact ledger, and an explicit evidence-class taxonomy. Its present archive supports software-path and affordance claims, not empirical validation of a population model. This distinction also responds to concerns that language models should not be treated as drop-in replacements for human participants (Dillion et al., 2023), particularly when generated outputs can flatten variation within identity groups (Wang et al., 2025).

3.5 Platforms, parallel execution, and counterfactual tooling

A number of recent platforms have approached parts of the artificial-society problem at production scale. Concordia (Vezhnevets et al., 2023) generalizes Generative Agents to a composable event-driven formulation with a focus on embodied conversational characters. OASIS (Yang et al., 2024) scales social-media-style simulation to one million agents. AgentSociety (Gao et al., 2025) adds a city-scale multi-agent environment with urban, economic, and information substrates. RecSim (Ie et al., 2019) and its successors approach recommendation and counterfactual evaluation from a different angle, treating each user as a model and each policy as an intervention. The research-engineering literature on parallel and distributed simulation (Fujimoto, 2016) emphasizes logical processes,

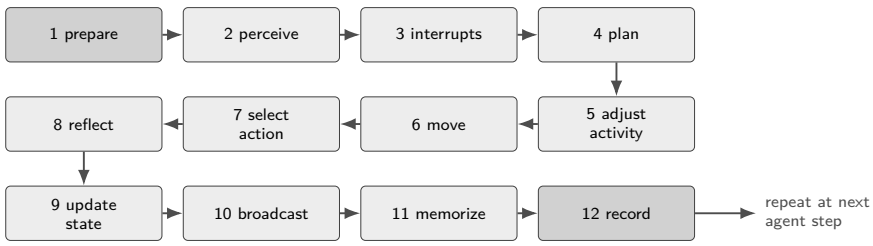


Figure 2: The configured default agent-step pipeline. The first and last stages are structural boundaries; intermediate stages may be omitted or replaced through configuration. Domain inputs also arrive through direct calls, event-bus contribution points, plugin lifecycle hooks, and a player intervention queue; the diagram is a control-flow account rather than a claim that every subsystem is a plugin.

conservative synchronization, and rollback. v2 does not adopt conservative synchronization: the hub-node protocol is a best-effort relay that allows local autonomy and tolerates brief disconnection, and the parallel-worlds engine is sequential at the world level, not at the agent level. v2 instead positions itself as a platform whose goals are configurability, auditability, and a single-cohort comparability across matched worlds.

The counterfactual-inference layer (paired ATE with bootstrap CI, sign-flip p, BH-q, pre-event balance / difference-in-differences, placebo noise floor, heterogeneity, dose–response, replication across seeds, and an optional one-call model reading) follows the parallel-worlds literature (Pearl et al., 2016; Hernán and Robins, 2020; Rubin, 2005) and the recent multiple-testing literature (Benjamini and Hochberg, 1995). It is deterministic and does not call a model during inference; the model call is reserved for the optional interpretation, which is itself bound to specific evidence rows.

4 Persistent and Situated Agent Lifecycle

GAWorld v2 represents an agent as a continuing participant in a shared simulation, not as an isolated chat session. The representation separates relatively stable profile material, mutable runtime state, and artifacts accumulated over simulated time. Figure 2 shows the connected per-tick control path, including v2's addition of a personality pluggable and the indoor-room resolution that occurs after movement.

4.1 Profile seeds, mutable state, and personality

At construction time, a tabular seed row supplies the identifier, demographic fields, and initial scalar state, while a Markdown profile supplies descriptive material such as occupation, daily life, and values. The current builder merges these sources into a dictionary containing `id`, profile fields, `state`, an in-memory `memory` list, `social_neighbors`, `personality` (Big-Five proxies), and `ext` (per-plugin namespaces); it then initializes location against the city map and personality against the configured personality plugin. The core state includes affective, socioeconomic, identity, behavioral, and information-exposure variables. Additional mechanisms may add namespaced or compatibility fields during setup and execution.

v2 adds a *personality plugin* that supplies Big-Five-derived proxies (openness, conscientiousness, extraversion, agreeableness, neuroticism) for every resident, used by the dynamic-interruption module and by selection probabilities across the platform. The plugin's anchors (a deterministic fallback) and a model-derived overlay (an LLM cache keyed by a profile signature) coexist; an experimental configuration can select the LLM overlay, the deterministic anchors, or a custom provider. The personality module also maintains a corpus-rewrite pass that compares anchor distributions across calibrators and warns when one trait dominates a group that the configuration does not intend to dominate, in the spirit of recent work on within-group flattening (Wang et al., 2025).

The class `gaworld.core.agent.Agent` is a typed adapter over the backing dictionary. It provides properties for identifier, name, state, memory, schedule, location, attached skill identifiers, and personality proxies, while dictionary access delegates to the same backing object. `view_as_agent` therefore introduces no copy, and `ensure_agent_dict` allows a typed caller to hand the record back to older helpers. This is important for

incremental migration, but it is not a fully typed agent schema: many runtime call sites still consume dictionaries, and arbitrary mechanism-specific fields can still be added.

4.2 Per-agent directory and persistence

Persistence extends beyond the in-memory record. Depending on enabled features, per-agent directories hold episodic entries, diaries, schedules, vector-indexed recall material, environmental preferences, private skills, organization memberships, and travel itineraries. On a stateful restart, stored memory is loaded into the agent and can seed retrieval. The file boundary makes inputs to later decisions inspectable, although it also means that reproducibility depends on capturing configuration, external model behavior, and the complete artifact directory rather than only the seed CSV.

4.3 The default per-tick path

The cognition loop is implemented as an ordered `StagePipeline`. A stage receives an agent, a shared per-step dictionary, and the `SimContext`. Hook-visible values retain names such as `activity`, `action`, `outcome`, and `reflection`; intermediate values use underscore-prefixed keys. The default path contains the following twelve stages, matching v1 with one explicit v2 addition:

1. **prepare** establishes tick-local inputs and emits the pre-step boundary;
2. **perceive** combines local environment, social, policy, memory, and contributed perception sections, including the indoor-room slot that names the room the agent is currently occupying;
3. **interrupts** considers events that can alter the scheduled course, drawing on the dynamic-behavior module and personality proxies;
4. **plan** forms the agent's immediate intention in the current context;
5. **adjust_activity** reconciles that plan with scheduled and interrupted activity;
6. **move** resolves travel and updates location, then resolves the indoor room that the activity places the agent in (the v2 indoor-room hook fires here);
7. **select_action** chooses an action from the available action space, optionally mediated by a Controller validator chain;
8. **reflect** produces an interpretation of action and outcome;
9. **update_state** applies state, relationship, and organization effects;
10. **broadcast** exposes eligible communication to other agents or a distributed relay, including a player-say channel for human participants;
11. **memorize** constructs and stores the tick's experience; and
12. **record** appends established logs, state history, Recorder events, and visual trace material.

Before the pipeline runs, the outer loop advances the shared Clock, updates the environment and local occupancy, polls distributed messages when enabled, and may perform configured information seeking. After it returns, the runtime emits `on_agent_post_step`, allowing mechanisms such as the intervention, governance, and organizations plugins to compute their own metrics. Day-start and day-end events bracket the tick loop; memory consolidation, governance tallying, and organization-rule resolution are invoked at the day boundary. The twelve stages are therefore the agent-step spine, not the entire simulation lifecycle.

4.4 Contribution points and compatibility calls

The EventBus lets a mechanism observe an event, contribute zero or more items, or filter a value in priority order. Perception composition is a concrete example: local context can be assembled by established code while plugins add bounded text sections without owning the full prompt. Post-step and day-end events similarly let policy, economy, skills, interests, organizations, or governance react at defined boundaries. Observer failures are logged and isolated unless strict mode is enabled; stage failures instead propagate because a missing control-flow stage could silently corrupt state.

Not all inputs use this bus. Memory recall, environment evolution, and social-network operations retain direct paths within or around the pipeline. The Controller now validates structured movement requests and exposes audited runtime interventions, but it does not mediate every selected action. Likewise, `record` writes established artifacts in addition to the kernel Recorder. The lifecycle is connected and configurable while retaining compatibility paths beyond the plugin boundary.

4.5 Player interventions and the away state

v2 adds two new first-class inputs to the lifecycle. A *player intervention* arrives through the same Controller queue as scheduled events; its type is one of `player_claim`, `player_act`, `player_say`, or `player_release`. In a single-process deployment, the intervention enters the local queue; in a multi-node deployment, the hub routes the intervention to the node that owns the resident, and the node enqueues it for the next tick. Either way, the intervention is then delivered to the agent's `interrupts` stage (claim/release affect the lease) or `perceive` stage (act / say appear in the next perception), and is recorded in the audit trail. The away state is a per-agent `travel` namespace that holds `status`, `purpose`, `place`, `distance_km`, `home_node`, `depart_day`, `return_day`, and `fare`; the agent's `locations.current` is rewritten to `away (place)` so that occupancy calculations exclude the agent without losing the round-trip. Coming home restores `locations.current` from `home_node` so that travel time and fare are computed against a real node rather than a label.

4.6 Configuration and ablation semantics

The stage order is read from `CONFIG["pipeline"]["agent_step"]`. Entries may name built-in stages or importable callables; a custom callable can also receive a display name. Unresolvable entries are skipped with a warning, and an empty resolved pipeline is rejected. Downstream built-ins generally read absent step values defensively, which supports mechanism ablations, but `prepare` and `record` are intended as structural boundaries. Changing order or removing a stage can change semantics and artifact completeness. The configuration surface enables controlled system experiments; it does not guarantee that every permutation is meaningful or behavior-preserving.

5 Microkernel and Plugin System

GAWorld v2 combines a connected cognition pipeline with a small kernel, a registry of fourteen built-in plugins, and compatibility-preserving domain paths. The K1–K5 migration audited in v1 is now closed for the defined scope, and a v2 audit covers three further migrations: *multi-tenant world resolution* (K6), *distributed hub-and-node execution* (K7), and *parallel-worlds counterfactual execution* (K8). The Figure 4 freezes the v2 architecture at the audited commit; existing capability cases are predate or postdate specific migrations and are labeled accordingly in section 12.

5.1 Kernel services

Simulation context and clock. `SimContext` holds configuration, Clock, EventBus, PluginRegistry, Controller, Recorder, agents and their identifier index, an LLM callable, a seeded random-number generator, an `extras` compatibility escape hatch, and (v2) a `WorldContext` reference to the active per-tenant world. It also provides simulation-level and per-agent plugin-state namespaces. The outer loop is the sole clock writer, giving stages, plugins, and records a common simulated day, time, and tick. This ordering does not by itself make model calls, external services, or concurrency deterministic.

Event bus and plugin registry. The EventBus supports priority-ordered observation (`emit`), merged contribution (`collect`), and value rewriting (`filter`). Handler failures are logged and isolated by default, so warnings and active plugin lists must accompany results. The PluginRegistry assembles built-ins, configuration-named classes, and installed entry points; orders setup by declared dependencies; skips failed or unsatisfied plugins; and tears active plugins down in reverse order. Plugins may own configuration, namespaced agent state, and output directories, but execute in process and are not a sandbox.

Controller. The Controller provides a priority-ordered validator chain over structured `ActionRequest`s and a named runtime-intervention API. In the main simulator, movement requests pass through validation: denial preserves the origin and is surfaced in the next perception, while a rewrite can change the destination. The standard runtime interventions set one agent-state value, update a dotted live-config path, queue agent removal for the next day boundary, and (v2) deliver a player intervention to a specific node's queue. Denials and interventions are audited through the Recorder.

Recorder. The Recorder appends clock-stamped JSON objects to table-specific files and is closed when the simulation exits. It records Controller activity, distributed-hub activity, player interventions, and is available to plugins. Main traces, state history, memories, environment events, and specialized metrics also retain established writers.

World resolver (K6). The world resolver turns a request's authenticated user, a cookie indicating the current world, and the caller's role into a `WorldContext` that selects the active per-tenant configuration overlay, output root, run state, and intervention queue. Path-resolving functions in the dashboard and simulation code consult the active world; module constants remain valid for the `default` world so that single-tenant deployments and unit tests behave identically. When the account store is absent, the resolver short-circuits to the `default` world.

Distributed hub (K7). The hub owns the per-world relay, the per-node intervention outbox, the node-token store, and the global Recorder merge point. The hub is the *dashboard server* when accounts are enabled and a thin `cluster/hub` when accounts are not. The hub does not advance simulation time; it serves the run's record and routes messages.

Parallel-worlds engine (K8). The parallel-worlds engine runs up to eight matched worlds under a shared seed and a per-world config patch, isolates each world's output tree, and collects per-arm completion status and per-resident trajectory diffs for post-hoc counterfactual inference. It reuses the same event-bus, Controller, and Recorder path; no domain code is changed between worlds.

5.2 Built-in plugin assembly

The audited registry assembles fourteen built-in plugins. They attach domain behavior to day/tick events, perception contributions, location filters, action or outcome filters, and post-step or day-end hooks. The following table summarizes their declared surface and an indicative per-agent or per-world state fragment.

The adapters delegate calculations and specialized persistence to established modules: migration changes orchestration and ownership, not the scientific validity of the mechanisms. The audit table in Table 2 lists which subsystems have been fully plugin-owned, which are partially integrated, and which retain direct compatibility paths.

5.3 Plugin authoring

A third-party plugin is a subclass of `gaworld.kernel.Plugin` with a declared `id`, a tuple of plugin `requires` dependencies, a `setup` method that receives `SimContext` and registers `ctx.bus.on(...)` handlers, and a `teardown` method. A plugin can be installed by configuration, by entry point, or by code registration (Project, 2026i). Configuration and entry-point installation are non-overlapping: a configured plugin runs even if it is not published as an entry point. Plugin setup failures are isolated: they produce a warning and the affected plugin is deactivated, but the rest of the registry continues to run. This is intentional and is the reason the plugin set can be reconfigured at runtime without an entire-process restart; it is also the reason that an active plugin list, with its warnings, is part of every research artifact.

5.4 Migration-state audit

5.5 Failure and reproducibility boundaries

A cognition-stage exception propagates because the stage is control flow. Bus-handler and plugin-setup failures instead warn and normally leave the remaining run active; missing dependencies deactivate the affected plugin. Operational resilience therefore makes a completed process different from a complete treatment. A reproducible artifact must include resolved stage order, active plugin identifiers, warnings, configuration, source snapshot, model/provider conditions, specialized outputs, the world identifier, the node identifier (if any), and the player intervention trail (if any). The v1 description of K1–K5 remains accurate; v2 adds the requirement that the artifact also record the world identifier, the node identifier, and the per-arm completion status of every parallel-worlds study. These additions are not yet enforced by the kernel; the artifact ledger remains the researcher's responsibility, with the schema documented in Section 13.

Table 1: The audited v2 plugin set. Per-agent state column lists the declarative fragment a researcher can read on a hashed run; the deployment column lists the post-v1 changes each plugin enables.

Plugin	Lifecycle hook	Per-agent state fragment	v2 addition
Intervention	on_day_start, post_step	`ext.intervention`	cross-tenant metrics
Skills	on_agent_post_step, memorize	private skill library	skill consolidation by personality
Interests	on_day_end, perceive	`ext.interests`	growth phase tracker
Life events	on_day_start, on_agent_post_step	`ext.life`	state effect normalization
Economy	on_day_end, on_agent_post_step	`state.econ`	endogenous inflation, credit-bound consumption
Local physical perception	perceive	local snapshot	indoor-room resolution
Real work	on_day_start, on_agent_post_step	`ext.work`	capability profile cache
Dynamic behavior	perceive, interrupts	`ext.dynamic`	personality-conditional interruption probability
Spatial preferences	filter (location)	`ext.spatial`	aversive-route rewrite
Personality	agents.built, perceive	`personality`	Big-Five anchors + LLM overlay
Indoor rooms and venue capacity	agent.moved, perceive	`locations.room_intent`	room-resolved occupancy and capacity
Organizations	on_day_start, on_day_end, on_agent_post_step	`ext.org`	persistent org with role
Governance	on_day_end, on_day_start	`ext.gov`	proposal/ballot engine
Offline citizens	agents.built	`ext.offline`	ghost taxonomy with visit / duty channel

6 Memory, Cognition, and Behavioral Realism

v2 separates persistence from generation. An agent's continuity is not held solely in an LLM context window: structured state, episodic records, habits, intentions, relationships, environmental preferences, growth profiles, private skills, organizational membership, governance state, travel itinerary, and personality proxies are written to mechanism-specific stores and reintroduced at later steps. Figure 6 summarizes this path. The system can preserve and expose information across simulated days; whether its memory dynamics resemble those of people is a separate empirical question.

6.1 Episodic persistence and retrieval

The basic store retains an agent's memory list, schedule, available actions, locations, and location–action biases as JSON, alongside append-only textual logs and simulation state. A second experience layer records episodes in JSONL and persists habits, daily intentions, relationships, and environmental preferences in separate per-agent files. This division supports inspection and selective restart, although a complete restart requires all of these stores rather than a single canonical agent file.

For content-addressed recall, memory entries are also indexed in SQLite with type, simulated time, salience, recall count, and an embedding. The configured embedding path may call an embedding provider; if that path is unavailable, the implementation falls back to a deterministic hash embedding. Retrieval combines vector similarity with salience, age decay, and prior recall, then optionally boosts entries related to the agent's current growth interests. Thus ranking is a specified scoring procedure over stored entries, whereas the text being ranked may have originated in an LLM-generated action, reflection, diary, news item, or external source.

The day-boundary lifecycle adds two optional transformations. Consolidation selects recent episodic rows and asks an LLM to distill them into a small set of semantic entries; malformed output or insufficient material produces no new entry. Decay is LLM-free: after a configured period it lowers the salience of old, unrecalled episodic rows

Table 2: Implementation state in the audited v2 repository. "Wired" denotes a subsystem whose boundary is now the plugin surface; "partial" denotes an adapter that contributes but retains direct calls; "retained compatibility" denotes a subsystem whose boundary is still the legacy direct call. K1–K5 cover the v1 migration; K6–K8 cover v2.

Surface	Current boundary	Status (K)	Retained path and consequence
Cognition	Configured 12-stage pipeline	Wired (K1)	Shared step dictionary; order and omissions must be reported
Extensions	EventBus and PluginRegistry	Wired (K1)	Failures may degrade rather than terminate a run
14 domain adapters	Policy, skills, interests, events, economy, physical, work, behavior, spatial, personality, rooms, organizations, governance, offline	Plugin-integrated (K1, K2)	Established stores and writers remain; activation needs config/plugin provenance
Memory	Recall, storage, consolidation, decay	Direct compatibility path (K3)	Persistent context is inspectable, not psychologically validated
Environment and social network	World evolution and network modules	Direct compatibility path (K3)	The full world and network models are not isolated plugins
Controller	Move validation, named interventions, player interventions	Connected, scoped (K4)	Other actions retain established paths; no universal-gate claim
Recorder	Clocked structured event writer	Connected, non-exclusive (K4)	Multiple writers remain; no single-log replay claim
World resolution	Cookie + account + role → WorldContext	Wired (K6)	When accounts are absent, single-tenant behaviour is preserved
Distributed hub	Per-world relay and per-node outbox	Wired (K7)	Best-effort delivery; no consensus, no clock synchronization
Parallel worlds	Up to 8 matched worlds, shared seed	Wired (K8)	Deterministic; no domain change between arms

and deletes rows already below a floor, while protecting semantic, external-information, and procedural entries. A separate episode store also computes exponential salience decay and enforces a size cap. These two decay paths operate over different stores and should not be mistaken for one unified cognitive theory.

External retrieval-augmented memory is also optional. v2 unifies three external-memory paths behind a single `info-sources` surface: a keyword-driven web-search RAG (Project, 2026k), an agent-derived query flow that retrieves web material, and a configured *information diet* that mixes social, news, and professional sources per agent. After retrieval an LLM may assess or compress the material into external-information entries. This path depends on a caller-supplied fetch adapter and degrades to no operation on failure. Consequently, an experimental report must disclose whether external retrieval, provider-based embeddings, consolidation, and decay were enabled.

6.2 Intentions, habits, and reactive behavior

Behavioral continuity combines model-mediated interpretation with explicit state-transition rules. At day boundaries, recent episodes and current state can be supplied to an LLM to produce priorities, avoidances, social targets, and recovery goals. When the call budget is exhausted or parsing fails, a rule-based fallback constructs intentions from state and recent experience. Habit strength, relationship signals, physiological and time-pressure needs, episode salience, and many mood or stress adjustments are updated by bounded numeric rules. These values enter subsequent prompt context rather than being left as unstructured prose.

The dynamic-behavior module evaluates candidate interruptions from needs, co-location, incoming messages, environmental events, local physical conditions, cascades, and spontaneous urges. Acceptance probabilities depend on event intensity, activity commitment, personality proxies, relationship closeness, and agent state. Candidate generation and schedule insertion are LLM-free but stochastic; they are repeatable only when the relevant random state is seeded. Once an interruption wins, the ordinary activity-adjustment or action path may invoke an LLM to phrase or refine the response. v2 therefore distinguishes rule-based opportunity and constraint generation from model-generated interpretation and action text, and adds a *personality-conditional* acceptance probability drawn from the Big-Five proxies the personality plugin supplies. The configuration can disable this overwrite and recover

the v1 behavior; the artifact ledger records which version was active.

Spatial preference illustrates continuity without language generation. A crowd surge or other location anomaly raises a time-bucketed aversion score; scores decay with recency, and a later destination can be redirected when its aversion crosses a configured threshold. This mechanism implements learned avoidance as a transparent rule. It does not establish that the learning rate, threshold, or decay reproduces human spatial choice.

6.3 Interests, skills, and productive activity

Growth profiles contain bounded hobby and skill items with phase, engagement, competence, practice history, and activity templates. Their initial content may be derived by an LLM and cached by a profile signature; deterministic fallback profiles are available when generation fails. Episode matching, practice updates, phase classification, daily retention decay, retirement of stale interests, and socially mediated adoption then use explicit rules and a seedable random generator. The labels borrow a developmental vocabulary, but their implementation is a simulation mechanism rather than a validated model of human interest development.

The skill subsystem maintains a curated global library and per-agent private skills. Available skills can be injected into cognition context, while an optional LLM consolidation step extracts a reusable procedure from recent episodes and saves it as a private Markdown skill. Parsing, schema checks, and registry precedence constrain this output; they do not demonstrate that the procedure is correct or transferable outside the generated setting.

Productive activity extends this pattern to a mock work market. Cached, LLM-derived capability profiles and current growth items inform routing. A JSONL queue provides claim and recovery semantics, a worker pool dispatches briefs concurrently, and code, content, teaching, and web-design adapters use an LLM to produce files. Lightweight validators check properties such as Python syntax, Markdown structure, or HTML/SVG parseability before successful results are ingested into agent state and market settlement. These checks and persisted artifact paths demonstrate a functioning production pipeline; they are not evidence that an artifact meets professional, pedagogical, factual, or security standards.

6.4 Realism module and personality anchoring

v2 exposes a *realism module* (`gaworld/cognition/realism.py`) that contains the heuristics used to make generated text more human-like without claiming a psychological theory. Concrete entries include non-cliché opening moves, time-of-day salutations, the calibration of mood to the day's events, and the suppression of repeated openings in successive diary entries. The realism module is consulted through `on_day_start` and `on_agent_post_step`; a configuration can disable it for a baseline run. v2 also adds a personality-anchor *rewriter* that, when invoked with an LLM overlay, calibrates Big-Five proxies against a corpus and warns when one trait dominates a group that the configuration does not intend to dominate. The rewriter is opt-in, and the artifacts that opt in record the calibration set, the resulting distributions, and a SHA-256 hash. The rewriter does not establish psychological validity; it is a controlled, inspectable mitigation of flattening, not a substitute for a held-out evaluation.

7 Situated Urban and Societal Substrate

GAWorld v2 places persistent agents in a shared substrate whose components read and write overlapping state. The city constrains movement, the indoor spatial tree constrains co-location, local occupancy changes perception, relationships affect encounters, financial or life events alter later choices, organizations impose rules, governance proposes and votes on rule changes, and information exposure shapes what residents believe. This coupling supports questions that cross individual and aggregate levels, but it also means that mechanism provenance and configuration are part of every experimental treatment.

7.1 City graph and indoor spatial tree

The city map parses places, roads, transit lines, interiors, and land-use attributes into a graph. Nodes carry categories, coordinates, capacity, opening hours, popularity, density, and display metadata. Destination resolution can match activity terms to place categories, while graph routing and transport-mode rules estimate distance, time, and fare. Mode speeds, fixed delays, road-class multipliers, rush-hour penalties, area price levels, and weather preferences are explicit parameters. They make travel costs inspectable; their presence does not imply transport-model calibration.

v2 extends the city graph with an *indoor spatial tree* that resolves every indoor activity to a specific room in a building. The tree shares its layout vocabulary with the front-end visualizer, so that back-end simulation and front-end display agree on which room an agent is in. Eighteen layouts (residential, apartment, hotel, hospital, clinic, pharmacy, school, library, shop, cafe, restaurant, office, government, factory, station, gym, park, and public space) map activity terms to furniture slots (sleep, cook, exercise, ...). When the spatial-tree plugin is enabled, indoor co-location restricts a meeting to the same household and the same room; venue capacity is per-room, computed as a fraction of the venue's capacity that the room occupies in its layout. When the plugin is disabled, the v1 behavior is preserved.

At each tick, the local-physical layer recomputes occupancy from stationary agents, clears stale counts, and exposes the current location's crowding, opening status, and weather. A crowd anomaly is flagged when occupancy is both high and sharply above the previous tick. These snapshots can enter perception and interruption generation, linking collective movement to individual behavior. The mechanism is deterministic given agent locations and parameters, although the locations themselves may follow stochastic or LLM-mediated decisions.

7.2 Environment, weather, and crowd anomalies

The broader environment system supports configured natural, economic, political, technological, and intraday events. In rule mode it samples from configured event distributions using a local random generator. In LLM mode a model proposes a daily summary, events, and intraday rules that are parsed into the same bounded event schema; configured fallback generation remains available if that call fails. Severity and anomaly tagging are rule-based. v2 separates *natural* events (weather, accident, epidemic) from *social* events (rumor, market shock, policy reform) and exposes a *crowd anomaly* channel that the local-physical layer writes into a single perception field. Reports must therefore identify the generator mode rather than treating all environmental histories as equivalent.

7.3 Social ties, life events, and offline citizens

Relationships are typed records rather than undifferentiated neighbor edges. Role templates specify channels, obligation, protection, and daily decay; interaction signals update closeness and contact time. The implementation can bootstrap off-screen contacts from an LLM-generated backstory, with a deterministic heuristic fallback. v2 adds an *offline citizen* subsystem that classifies each ghost contact by relationship role and admission channel (call, message, visit, none), produces stateful obligations when a contact is long-neglected, and exposes a visit channel that the *travel* plugin (Project, 2026m) actually exercises. The combined offline + travel stack turns a previously read-only field (the obligation counter) into a behaviorally consequential signal.

Maintenance is rule-based. Uncontacted ties decay at role-specific rates, long-neglected obligations can rise, and an enforcement pass assigns nested Dunbar-style tiers while pruning beyond a configured outer cap and protecting selected kin roles. These tiers are an operational capacity constraint, not evidence that the simulated network reproduces measured human networks.

Life events are normalized, persisted, scheduled by simulated day and time, and drained once when due. Events can target agents and carry bounded state effects and tags. v2 makes life-event state effects typed and reversible: each effect has an effect-id, a magnitude, a half-life, and a reversal hook, so a cancelled event can be undone at the day boundary without leaving free-floating state. This is a controlled exogenous-event surface whose timing and direct updates are inspectable; the realism of event frequencies, dependencies, and consequences must be established by a separate study.

7.4 Economy, credit, and endogenous inflation

The finance module maintains checking, savings, investment, and housing-fund accounts; monthly budgets; income and expense records; liabilities and credit; sector pools; government flows; and macro state. Configured rules calculate social-insurance deductions and progressive income tax, route wages and consumption, enforce available-cash and credit constraints, transfer savings, and settle payments. A dedicated seeded random generator drives expenditure variation, market returns, shocks, and other probabilistic transitions so that unrelated random draws need not perturb the economy sequence.

v2 closes a long-standing gap: the v1 code reported an inflation rate and a price index that no other component read. In v2 the price level multiplies real expenditure so that wages that do not follow it produce a measurable erosion of real consumption. Inflation itself is endogenized through a *demand-pull* rule: $\pi = \pi^* + \kappa \cdot (C/C_0 - 1)$, where C is a 30-day moving average of real per-capita consumption, C_0 is the first 30 days as a baseline, and π is

clamped to a configured interval (Project, 2026e). The wage index, when enabled, ties a fraction of new wages to the price level. This is the first run at making "cost-of-living erosion" actually measurable in a GAWorld run, and the artifact ledger records whether the wage index was active so that the directional claim is not silently over-interpreted.

Shared sector and government pools make counterparties visible instead of creating every monetary flow without an account. Agent-to-agent payments and some rent or labor flows can be routed to modeled recipients; investment returns combine a common market component with idiosyncratic variation. Ledger and conservation checks can reveal implementation defects such as unbalanced internal transfers. They cannot establish that the tax schedule, consumption response, labor process, credit behavior, or macro dynamics is an empirically valid representation of a real economy. Accounting consistency is therefore a software invariant, not a behavioral-validation result.

7.5 Organizations, governance, and rule changes

v2 adds a persistent-organization subsystem. Each organization is a typed object (community, business, club) with members, role templates, a chosen governance mode (*off*, *leader*, *member-vote*), and a ruleset that the plugin applies on member decisions. Two rule-change pipelines exist: a *leader* pipeline in which a designated member decides, and a *member-vote* pipeline in which a proposal is opened, voted on, and (if it passes) enqueued for application at the next day boundary (Project, 2026d). Generations are isolated: a proposal opened under a given member snapshot, governance version, and ruleset cannot be re-opened after the world is rebuilt with a different member snapshot. This isolation is enforced in a single SQLite transaction.

The governance subsystem is opt-in (`governance.enabled`); when it is off, the v1 organization behavior is preserved. Vote tallies and rule versions are recorded in a separate `governance.json` per generation, not interleaved with the per-step metrics CSVs, so that governance results do not contaminate aggregate economic measurements.

7.6 Information sources and Moltbook

v2 unifies information exposure under an *info-sources* surface (Project, 2026k). Each resident has a configured diet (a list of channels and a sampling rate) and a per-tick search budget. Channels cover local news, social posts, professional feeds, and (when enabled) web search. The Moltbook integration gives residents a social network identity (`gaworld/moltbook`) where they post, reply, and react; the agent's actions in Moltbook are recorded as separate `log.jsonl` entries, distinct from simulation traces, and are summarized back into the perception field at the next tick. This separation lets a researcher inspect the social-network behavior of a resident without conflating it with the simulation's per-step state history.

7.7 Coupling and ownership boundaries

Table 3 makes the principal shared-state paths explicit. "Implemented" refers to an inspectable current mechanism; "partial boundary" indicates that the mechanism is connected but retains direct calls, shared dictionaries, or specialized writers rather than a fully isolated plugin interface.

The table exposes the current modularity limit. The 14 built-in plugins have plugin entry points; economy, local physical perception, spatial preferences, work, dynamic behavior, indoor rooms, organizations, governance, personality, offline citizens, and information sources have built-in adapters. Memory, environment evolution, and social-network operations still exchange mutable agent dictionaries or runtime objects directly. This arrangement is usable and inspectable, yet replacing one mechanism can change prompts, state schemas, random draws, and specialized artifacts elsewhere. Controlled comparisons should therefore freeze the complete coupled configuration rather than naming only the focal intervention.

8 Multi-User, Multi-World, and Per-Tenant Isolation

GAWorld v2 makes a *World* a first-class object. A world is a city bundle plus a per-world configuration overlay plus a per-world output root plus a per-world set of memberships. The active world is resolved from a cookie, validated against the caller's role, and propagated through a context variable. This section describes the data model, the request-resolution path, the run-management machinery, the visibility rules, the static-file guard, and the per-tenant quotas. When the account store is absent, the single `default` world is the only world, and the system behaves as it did before multi-tenancy was added.

8.1 Data model

A v2 deployment, when accounts are enabled, stores the following objects in an SQLite database (`output/accounts/accounts.sql` no new dependency):

- **User** with one of two roles: *admin* (teacher) or *member* (student), and an optional capability bit `can_create_city`;
- **Session** storing only a SHA-256 hash of a 32-byte random token, with `HttpOnly; SameSite=Strict` cookies;
- **Invite** generated in batches, optionally labelled, with an expiry and a quota template;
- **World** referencing a city bundle, a per-world configuration file, and a visibility level;
- **Membership** binding a user to a world as *owner*/*player*/*observer*;
- **Usage** and **audit** records for per-tenant quotas and forensic replay.

A world is a folder under `output/worlds/world_id/` that holds the city's `agents.csv` and `profiles.md` as copy-on-write `seed/` files, the run log, the run-time configuration overlay, the run state, the intervention queue, and the Recorder. When the world is deleted, the folder is removed; when the user account is deleted, every owned world is deleted with it. A world is also the natural unit of distributed execution (see Section 9).

8.2 Request resolution

A request enters the dashboard server with a session cookie, a `gaworld_world` cookie indicating the caller's chosen world, and (optionally) a Bearer token. The `_guard` middleware authenticates the caller and loads the membership. The world resolver then:

1. reads the caller's `gaworld_world` cookie;
2. verifies that the world exists and that the caller has at least *observer* visibility;
3. if not, falls back to the shared `default` world (a per-tenant deployment may have no such world, in which case the request fails closed);
4. loads the world's `config.json` and merges it with the global `dashboard_config.json` under a documented precedence order;
5. sets a `ContextVar` that downstream code consults in lieu of reading global configuration; and
6. selects a path-resolving function for the world's output root.

This pattern preserves all module-level path constants in the v1 code: `output/` is still the default world's root, and the same code that reads a relative path in a single-tenant deployment reads a world-rooted path in a multi-tenant deployment. Unit tests that monkey-patch module constants continue to work; the dashboard does not require a new fork because of multi-tenancy. A pitfall identified during the migration is that `python -m gaworld.apps.dashboard_server` previously loaded the module a second time as `__main__`, producing two parallel world resolvers. The fix was to delegate the `-m` entry point to the canonical module and to add a regression test (`ModuleEntryPointTest`).

8.3 Path resolution and output isolation

Without multi-tenancy, the simulation wrote to a fixed set of paths under `output/`. v2 adds a path-resolution layer (`world_paths`) that takes a base root and returns the per-world versions of each path. The base layer is unchanged for the default world, so single-tenant deployments see no behavior change. For a multi-tenant deployment, the path layer reads `effective_config` and `world_root` from the context variable and rewrites the `RUN_PATHS` entries (`memory_dir`, `state_output_dir`, and `kernel.interventions_path` for the simulation, plus `records.output_dir` and `records.interventions_path` for the dashboard) to the world's root.

A run is launched in the world's process group with `GAWORLD_CONFIG_OVERRIDES` carrying the world-specific paths. The simulation code does not need to know which world it is in; it reads the override environment variable and writes to the resolved paths. A configuration change in a world's `config.json` therefore takes effect at the next run boundary without restarting the dashboard.

8.4 Visibility and roles

A world's visibility is one of *private*, *class*, or *open*. A *private* world is visible only to its owner (and any admin). A *class* world is read-only for all members; teachers use it to publish a demonstration world. An *open* world allows any member to join; joining creates a *player* membership that lets them claim residents and act in the simulation. A *player* can also be downgraded to an *observer*; an *observer* can read state, profiles, and memory but cannot mutate anything.

Roles are enforced at the API layer, not at the run layer: the run itself is unaware of who is watching, but a write to the world is checked against the caller's role before being applied. This is the same trust model as the v1 system but with a finer-grained permission set.

8.5 Run management and quotas

A world can have at most one running simulation at a time; the dashboard also enforces two global limits configurable by the admin:

- `max_concurrent_runs` (default 4): the maximum number of simulations across all worlds in the deployment;
- `max_runs_per_user` (default 1): the maximum number of concurrent simulations per user (admins are exempt).

A run request that exceeds a limit is enqueued in a FIFO queue whose head is the run with the earliest request time; the dashboard reports "queued (position k)" in the panel. A user can withdraw a queued request without affecting other users. Quotas and queue state persist across dashboard restarts.

8.6 Static-file guard

v2 introduces a static-file guard that requires an authenticated session cookie to read any path under `output/` or `data/`. The v1 implementation had only a single-token guard and rejected paths whose basename began with a dot; this was insufficient because a logged-in user could read other worlds' output. The v2 guard checks the world identifier, the caller's role, and the path's owner world before returning the file. Static assets outside `output/` and `data/` (CSS, JS, fonts, HTML) are unaffected.

8.7 Per-tenant ledger and audit

Every mutation that crosses the API is recorded in the `audit` table, with the caller's user id, the world's id, the operation type, the target entity, and a timestamp. The audit table is small (one row per write) and is intended for forensic replay rather than as a primary trace. A per-tenant `usage` table records the number of LLM calls the tenant has made, broken down by model, by task, and by day. The quota system reads from `usage` and refuses to start a run that would push the tenant over the daily quota; a configurable dry-run mode warns without refusing.

8.8 Backwards compatibility and migration

The migration to v2 is, by design, backwards compatible. The `AccountStore` is optional: an installation that does not run `python -m gaworld.accounts init` continues to behave exactly as v1 did, with `GAWORLD_DASHBOARD_TOKEN` as the only credential. A first admin can be created with the same command; from that point on, `_guard` requires a session cookie or a Bearer token, and `GAWORLD_DASHBOARD_TOKEN` is reserved for scripts and back-compat. A migration script exists for installations that want to adopt multi-tenancy without re-creating their world, but most production deployments start fresh. This is the only way to preserve the byte-for-byte single-tenant behavior that v1 promises; v2's promise is that single-tenant behavior is preserved exactly when the account store is absent.

9 Distributed Worlds and Playable Cities

v2 introduces a hub-and-node protocol that lets a single open world be partitioned across many machines. The hub is the dashboard server; each node is a worker machine that runs the simulation for a subset of the world's residents with the node's own model key. Players continue to interact through a browser; the hub routes their interventions to the node that owns the relevant resident. Figure 9 depicts the topology.

9.1 Topology and threat model

The topology is asymmetric on purpose: all traffic is node → hub. Nodes can sit behind NAT, behind phone hotspots, or in a firewall; they need only an outbound HTTPS connection to the hub. This is a deliberate trade-off: the protocol does not need a clock-synchronization primitive because the hub does not advance simulation time. Each node runs the simulation locally; the hub merges per-node Recorder rows and routes interventions. No transactional consistency is offered; the protocol is best-effort and tolerates brief disconnection.

The hub is the dashboard server; this means one port, one reverse proxy, one TLS certificate for a public deployment. The v1 standalone relay (`distributed_comm_server`) is retained for OpenClaw-style deployments, but

in v2 it is no longer required: the dashboard server embeds the relay as an in-process `DistributedRelayBackend`, one per world. The hub itself is also a node, identified as `node_id = hub`; it runs the simulation for residents that have not been assigned to any external node, so that the world has at least one heartbeat even when every external node is offline.

9.2 Node lifecycle

A node is added by the world's owner through the dashboard. The owner fills in a name (e.g., "Lab 302 seat 7") and a list of residents that the node should run (either a list of ids or "rebalance N"). The hub issues a one-time token of the form `world_id.node_id.secret`, shows it once, and stores the SHA-256 of the secret. The secret is therefore a bearer token; its revocation is immediate, and a revoked node receives a 401 on its next heartbeat and stops the simulation.

The node process is a single Python command:

```
python -m gaworld.cluster join <hub-url> --token <token>
```

The node begins by sending a heartbeat. The hub responds with "world is running; new run id is `R`; please download the world bundle". The node downloads a zip containing the world's configuration, the seed `agents.csv` and `profiles.md`, and the city bundle. It then launches the simulation subprocess with `GAWORLD_CONFIG_OVERRIDES` carrying the world's configuration, the per-node paths, the assigned residents, and the relay's URL and token. The simulation code does not change between local and remote modes; only the override environment differs.

While the simulation runs, the node polls interventions every second, enqueues them locally through the standard `kernel/remote` interface, and the simulation drains the queue at the next tick as in v1. The node also reads the per-table `records/*.jsonl` files every second and uploads new rows to the hub, which appends them to the world's merged Recorder with a `_node` tag on each row. When the hub says "stop", the node terminates the subprocess; when the subprocess exits on its own, the node reports that status in the next heartbeat. When the network is offline, the node keeps running and back-fills records when the connection returns; when the simulation's tick-sync timeout expires, the node advances without waiting.

9.3 Player action routing

A player intervention (`player_claim`, `player_act`, `player_say`, or `player_release`) is received by the hub through the dashboard's `play` API. The hub resolves the intervention's target resident and routes it:

- If the resident is on the hub, the intervention is appended to the hub's local queue (as in v1).
- If the resident is on a remote node, the intervention is appended to that node's outbox; the next time the node polls (<1 s in the v2 implementation), the intervention enters its local queue and is consumed at the next tick.
- For `player_say`, the speaker and the listener may be on different nodes. The speaker's node records "you said X to Y" as a local action; the listener's node receives a `player_hear` intervention that adds "Y said: X" to the listener's next perception.

Player leases remain on the hub (`play_api._CLAIMS`). A node's simulation does not need to know that the player exists; the lease is the hub's bookkeeping, refreshed every 30 s. If a node disconnects, the lease remains valid on the hub and the player can continue to be served; when the node reconnects, it re-registers any active claims so the simulation can keep them consistent.

9.4 Clock synchronization

The hub and each node maintain a *logical* clock that is updated whenever a Recorder row arrives. The simulation is allowed to advance its own clock without coordination, so the protocol is not a conservative-synchronization protocol: it tolerates drift. Each node periodically reports its clock to the hub, and the hub publishes a *median* clock that downstream code can use to align downstream consumers (the visualizer, the Auditor UI, the intervention display). The current implementation does not implement rollback or deterministic-replay, and the protocol does not promise a tighter-than-seconds clock alignment. This is the trade-off that makes the system operable behind NATs and is the reason the protocol is best-effort.

9.5 Security and access control

Node tokens grant access to the per-world `/api/cluster/node/*` and `/api/cluster/relay/*` endpoints; they do not grant access to the dashboard, the account store, or to any other world. Bearer authentication is required for every request; the node id is bound to the secret, and the secret is checked against a SHA-256 hash. The hub emits a structured audit row on every node request, including "heartbeat", "bundle-download", "intervention-poll", "records-upload", "sync", and "denied". A revoked node receives 401 on its next request and stops the simulation. When a node's `node_id` is reused after a restart, the new secret replaces the old hash; the old secret is invalidated atomically.

9.6 Deployment profiles

A v2 deployment can be configured in three profiles:

- **Single tenant, single process** (v1-equivalent): accounts are off, the relay is in-process, and the only node is the hub. This is the default for local research and for the bundled benchmarks.
- **Multi-tenant, single process**: accounts are on, the relay is in-process, and the only node is the hub. This is the default for a classroom or workshop.
- **Multi-tenant, multi-node**: accounts are on, the relay is in-process, and one or more external nodes are connected. This is the default for cross-machine deployments (a school lab, a cross-campus collaboration, a multi-region test).

The same code path runs in all three profiles; the differences are configuration, not code. A multi-node deployment does not promise horizontal scalability, throughput improvement, or fault tolerance; it promises the ability to assign residents to machines that have their own model keys, and the ability to keep a world alive when one node disconnects.

10 Research and Counterfactual Workflow

GAWorld v2 treats an experiment as a configured execution that produces an inspectable artifact tree, with each numeric result bound to a hash and an evidence class. The platform now exposes four research surfaces on top of the underlying simulation: a matched-event workflow inherited from v1; a parallel-worlds counterfactual engine; a Research Workbench that compiles a paper or a research idea into a pre-registered protocol; and a policy-simulation-and-optimization loop. Each surface produces its own artifact set and binds every result to the producing code path. This section describes the four surfaces; the Agent Playground and Arena are covered separately in section 11.

10.1 Matched event and baseline runs

For intervention studies, the `compare-event` command constructs two scenario directories under a timestamped comparison root. The `without_event` arm inherits the configured policy-event list, whereas the `with_event` arm appends a named event at an explicit simulation day and clock time. Both arms receive the same requested agent subset, simulation horizon, random seed, and—when supplied—LLM provider override. Before execution, each arm is reset within its own isolated memory, log, vector-database, state, network, environment, intervention, and visualization directories. The two runs are then launched concurrently. This organization reduces accidental cross-arm state leakage and makes missing or failed arms visible through separate reset and run logs.

After both subprocesses finish, the comparison routine reads each arm's `state/agent_state_history.csv`. For every available metric, it computes the across-agent final snapshot, the mean over recorded observations, and event-minus-baseline differences. Machine-readable results are stored in `comparison_metrics.csv`; `comparison_summary.md` provides a human-readable view. The separate `run_meta.json` records choices such as seed, horizon, provider, world identifier, node identifier, and whether the lower-fidelity fast mode was requested. v2 adds a `warnings.json` that records any plugin or stage that failed to load and the active plugin list; this makes the artifact self-describing about which mechanisms were actually present.

10.2 Parallel worlds and counterfactual inference

The parallel-worlds engine generalizes the matched-event workflow. An *experiment specification* describes up to eight worlds, a shared seed, a per-world `config` patch, a per-world event list, and (optionally) a set of

replication seeds. The engine runs the worlds sequentially under a shared `GAWORLD_CONFIG_OVERRIDES` skeleton; the seed and the patch together determine the per-world trajectory. Per-arm completion status is recorded in `worlds/<id>/run_meta.json` and the per-arm artifact trees are kept separate so that one arm's failure cannot contaminate another.

After all arms complete, the inference layer (`parallel/causal.py`) computes a deterministic counterfactual summary from the recorded state trajectories. No model call is made during inference. The summary includes:

- **Paired per-resident ATE** with a bootstrap confidence interval and a sign-flip p-value, for every metric that both arms recorded;
- **Pre-event balance** and a difference-in-differences estimator that uses the matched pre-event window as a baseline;
- **Placebo noise floor**: a separate placebo comparison that shows how large a difference arises from no intervention at all, with the same seed;
- **Onset order** and **heterogeneity**: the step at which the two trajectories first diverge and the per-resident effect size distribution;
- **Dose–response** for design points that vary an intervention parameter across worlds;
- **Replication** across seeds, with a per-seed ATE and a meta-analytic mean; and
- **Multiple-testing correction** with Benjamini–Hochberg q-values.

A study verdict is a function of the deterministic summary: a hypothesis is *supported* if its signed effect has $q < 0.05$ and the replication mean agrees in sign; *reversed* if the sign agrees but the effect is negligible; *undetermined* if the replication is mixed; and *unmeasured* if the producing metric is missing in either arm. The verdict gate is part of the same code path that the v1 GAWorld-Bench used, so the meaning of a verdict is consistent across the two surfaces.

An optional one-call model interpretation (`parallel/interpret.py`) follows the inference. The model sees the inference summary and the per-comparison verdict and produces a brief narrative; every claim in the narrative is bound to a specific evidence row and a verdict, and the interpretation is cached per comparison so that reruns are deterministic modulo provider noise. v2 does not allow the model to overturn a deterministic verdict: the verdict is the verdict.

10.3 Research Workbench: from a paper or idea to a protocol

The Research Workbench (Project, 2026c) divides a study into a planning phase (output: a plan document) and an execution phase (output: a verdict and a report). The planning phase is a *plan generator* that takes either a research idea in free text or a paper (URL or PDF) and emits a *plan* with a theory, a methods paragraph, key variables, hypotheses, a feature map to the GAWorld surface, two or three experimental designs, an implementation outline, a credibility plan, and a list of limits. The plan is editable; the model cannot write to a feature that is not in `FEATURES.md`, and the protocol compiler cannot use a metric that is not in the *measurable catalog*.

The execution phase is a *protocol compiler* that takes the chosen design and produces a *pre-registered protocol*: a list of conditions, a list of outcomes with pre-specified effect directions and minimum effect sizes, a list of pre-registered exclusion rules, and a list of confidence-thresholds. A pre-execution check (`research/evaluate.py`) verifies the protocol against the measurable catalog and refuses to start a study that fails the check. After the study runs, the deterministic verdicts are compared with the pre-registered expectations; an *unexpected* verdict is reported as such and is not rewritten to look like the prediction. The final report contains the pre-registered protocol first, then the results, then the interpretation, in that order. This pattern follows the spirit of the recent pre-registration literature in computational social science and is intended to discourage post-hoc rewriting.

10.4 Policy simulation and optimization

The policy-simulation loop (Project, 2026j) takes a description of a policy (and, optionally, a candidate version) and a fixed-scale sample of residents, and produces a JSON reaction per resident. The deterministic scoring aggregator converts the residents' JSON into a composite score. v2 then optionally calls a model with the score and the policy to propose modifications; the modified policy is re-simulated on the same resident sample, and the policy with the highest composite score is the recommendation. This is an *optimization* in the sense that the loop produces a candidate that scored higher than the original on the deterministic composite; it is not a guarantee that the higher score reflects a better real-world policy. The loop is bounded by a maximum number of iterations and by a hard cap on the number of LLM calls; the artifact records both budgets and the policies it produced.

10.5 Serious games and workshop studies

The *serious-game* module (Project, 2026l) is a turn-based role-play surface in which two or more roles are seated by a resident agent or a human through a seat link. Each round, the seated participants respond simultaneously; a facilitator call resolves the round; a debrief scores each role on a fixed rubric. Designs and sessions are stored under `output/research/serious_games/` and re-loadable for inspection. The serious-game surface is intended for participatory studies in which the researcher wants to see how a human participant negotiates with simulated residents under a fixed scenario; it is not a high-throughput evaluation surface and is not used for benchmark scoring.

10.6 Group interviews and persona distillation

A *group interview* is a one-shot elicitation across a roster of residents, run as a child process per city and aggregated in the dashboard (Project, 2026f). Each resident answers a fixed set of questions and the result is written to `output/interviews/<session_id>/` together with the resident's response, the producing code path, and the model and seed that produced the answer. A *persona distillation* takes a name or a URL and produces a sourced portrait (identity, state, Big-Five, mental models, expression DNA, limits) that can be deployed as a resident; the portrait and the producing research notes are stored under `output/personas/<slug>/` and only *deploy* writes the seed CSV (Project, 2026g). The deploy step is a separate, reviewed action that the system refuses to perform automatically, so that an unverified portrait cannot enter a population.

10.7 What the workflow identifies

The four surfaces above are useful scaffolds for counterfactual experiments, but the recorded differences are still single contrasts produced by an LLM-mediated dynamical system. A common seed does not guarantee common latent randomness after prompts diverge, and a policy description does not guarantee that an economic or mobility mechanism changed. Before interpreting a difference, a researcher should verify the intervention's path from event to structural state, replicate both arms over seeds, pre-register target metrics, and include placebo events or negative-control outcomes (Lipsitch et al., 2010). External policy conclusions additionally require a defensible mapping to the population and institutions of interest. GAWorld v2 currently supports constructing and auditing such a study, not replacing it. The deterministic verdict gate and the per-arm completion audit make a defensible study easier to perform than the v1 system made it, but they do not make a non-defensible study a defensible one.

11 Agent Playground and Arena

Beyond the research surfaces, v2 ships two evaluation surfaces that probe residents on a smaller scale. The Agent Playground is a library of fourteen game-based probes; the Agent Arena is a benchmark harness. Both surfaces are *in-memory* by default, write only their configured archives, and never modify a city bundle. They are designed to make a small group of residents answer a structured question in a way that can be inspected, replayed, and scored under deterministic rules.

11.1 The fourteen game families

The Agent Playground (Project, 2026h) covers four broad families, grouped by the kind of evidence the probe produces. Each game has a deterministic scoring rule that does not depend on a judge's LLM call, and a *judge* layer for the games whose scoring is genuinely qualitative. The four families are:

- **One-on-one**. *Persuasion*: a researcher asks a resident a question, has a brief conversation, and re-asks; the judge model compares the first and last answer to decide whether the resident changed their mind. *Read the room*: a resident is given a moral dilemma and asked to predict what another named resident would do; the scoring rule is exact match to the named resident's actual response. *Jury*: three to five residents hold a three-round discussion, vote anonymously, and decide on a sentence; the verdict is the majority, not the judge model.
- **One group, one event**. *Disaster mode*: a group of residents passes through a disaster stage by stage; each resident produces a structured JSON (action, panic, help, quote); the histogram is deterministic. *Warzone*: when a warehouse runs low, residents propose and vote on a daily basis; mortality, the Gini coefficient, and the

vote-disagreement rate are all deterministic. *Rumor spread*: a rumor is seeded, the model picks a verb (forward / verify / refute / ignore), the graph picks a recipient, and the run produces a propagation tree. *Referendum*: a self-interested bill is voted on privately and then publicly; an optional propaganda paragraph can be inserted; the shift in public opinion is reported. *News commentary*: a news item is shown to one to twelve residents who each produce a single comment with a stance and an intensity.

- **Mission / creative**. *Team duel*: two teams work on the same task from opposing angles; reviewers blinded to team name score on dimensions. *Novel writer*: two to six residents become protagonists of a single novel with an outline.
- **Long-form / human-in-the-loop**. *Roommate mode*: two to six residents share a virtual apartment; the front-end polls the simulation each tick. *Who's human*: residents and a real human sit in an anonymous group chat; the human enters through a seat link. *Timeslip*: two to eight residents are dropped into a real dynasty's decisive moment; each round produces an era-styled line and a modern inner monologue; a three-axis rubric (era fit, stability, inner conflict) is deterministic.

This is a heterogeneous set. Some games are deterministic given the residents' responses (*warzone*, *rumor*, *referendum*, *disaster mode* for its histogram), some depend on a judge LLM (*persuasion*, *team duel*, *novel writer*), and some are genuinely participatory (*who's human*, *roommate mode*). The common interface is a fixed rule for routing, scheduling, and archiving; the scoring rules differ and are documented per game.

11.2 Archives and the Track B scorecard

A subset of games (*rumor spread*, *referendum*, *disaster mode*) writes an archive to `output/games/<game>/<timestamp>-<job_id>` on completion. The archive contains the per-resident response, the round-by-round state (if any), the deterministic scoring result, the model that produced each response, and the user that initiated the run (in a multi-tenant deployment). These archives are the data source for the *GAWorld-Bench Track B* scorecard, which checks three pre-registered laws ("referendum conformity", "disaster solidarity > panic", "rumor decline after refutation, but not to zero"). Each law is tested with at least five runs; if fewer than five usable runs are available, the law is reported as *withheld* and not as *supported* or *reversed*. Track B is therefore a structured way to combine across-game evidence under a pre-registered rubric, distinct from Track A which is a software-path diagnostic.

11.3 Agent Arena

The Agent Arena is a benchmark harness for a fixed set of residents (Project, 2026b). The arena takes a target city, a set of residents (up to twelve), and a question set. A question set is either a built-in bank of ten questions covering math, QA, logic, reading, and translation, or an LLM-generated question set of arbitrary size, difficulty, and category. Each resident answers each question; the score is correctness and median latency. A researcher can apply a Top-K retention policy and the residents outside the top K are marked as *retired*; the researcher can also pull new residents from another city or from a batch-import file to replace the retired residents.

The arena is *not* a population benchmark. It does not produce a statement about the model's overall ability, only about a small set of named residents under a specific question bank. This restriction is deliberate: residents in GAWorld are not interchangeable with each other, and a benchmark that mixed their identities would not produce interpretable evidence. The arena's archive is per-resident and per-question, and it is the researcher's responsibility to compare results across cities or across model configurations in a defensible way.

11.4 State and ownership

A game session is held in memory; the dashboard exposes a "recent sessions" panel that displays the last twenty sessions. The `JobStore` for game sessions is per-game, in-memory, and is discarded when the dashboard restarts. Archives, by contrast, are persisted to `output/games/<game>/` and survive restarts. In a multi-tenant deployment, every archive is stamped with the world identifier and the user that initiated the run. Game state is not part of the city bundle and is never written to it; the only side effect on the bundle is the existence of an agent (and that effect comes from the underlying city, not from the game).

11.5 What the games evaluate

The Agent Playground is a probe layer, not a population benchmark. Its value is in producing per-game archives that can be inspected and compared across runs, and in providing a small set of pre-registered laws that the Track

B scorecard tests. The games do not establish that the residents are realistic, that the underlying model is a good representative of a population, or that the scoring rules are calibrated against human behavior. A finding in Track B is therefore a hypothesis about how the residents behave under a structured scenario, not a generalization to a real population. This is the same validation posture as the v1 system paper's Track A: the harness is a software artifact, the evidence it produces is a per-game archive, and the verdict is a deterministic function of the archive.

12 Existing Capability Cases

This section audits what the archived v2 outputs demonstrate about GAWorld as a research instrument. It does not treat an available file as a successful scientific evaluation. We use the v2 evidence classes: REAL for a usable GAWorld run artifact; DIAGNOSTIC for a re-analysis of a REAL artifact; SYNTHETIC for a fixture produced by `--synthetic`; INCOMPLETE for a missing, interrupted, or insufficient output; DESIGNED for a protocol that exists without usable outcome evidence; and FIXTURE for synthetic data that tests analysis code. These labels describe evidence availability, not system performance.

12.1 Case 1: Persistence is observable, but the memory contrast is not

The memory-consistency archive is useful because it exposes a longitudinal experimental shape that is often hidden in agent demonstrations. One agent (ID 34) was assigned to four intended treatments: intact memory, reset between phases, selective summary deletion, and injected conflict. Each arm planned a seven-day acquisition phase followed by a seven-day continuation phase. The archive contains seven phase-1 diaries for every arm, together with state and memory files. It also records different memory-file footprints, including episodic, growth, habit, and relationship artifacts. Thus a researcher can inspect what persisted, what was deleted, and how a later elicited action was represented.

Only the intact-memory arm, however, contains all seven phase-2 diaries. The reset, selective, and conflict arms each stop after phase 1, with the archived report attributing the interruption to initialization after the worker pool started. The archive therefore demonstrates file-backed persistence across two phases for one agent in one arm; it does not compare the behavioral effect of memory retention, deletion, or conflict. Qualitative labels such as a move from "procrastinating" to "active" are post-hoc codings of a few generated actions, not a validated continuity measure. In particular, differences in phase-1 final actions cannot explain missing phase-2 outcomes. This failure is scientifically informative at the infrastructure level: treatment completeness must be checked before interpreting a fluent diary as longitudinal evidence.

12.2 Case 2: Policy artifacts exercise a counterfactual workflow

The v1 policy archive contains four timestamped comparisons—traffic restriction, housing subsidy, medical reimbursement, and job-training subsidy. Each stores an event arm beside a baseline arm and exports the same metric columns, thereby exercising the isolation and reduction path described in section 10. All are single archived comparisons, with events injected on simulation day 3 (09:00 for traffic and 08:00 for the other three); there is no multi-seed sampling distribution or uncertainty estimate.

The raw CSV files also show why mechanistic inspection must precede substantive interpretation. In every pair, the final values of `cross_viewpoint_exposure`, `misinformation_risk`, `stance_score`, and `toxicity_score` are zero in both arms. Some generic state variables differ: for example, the housing comparison has a final event-minus-baseline `time_pressure` difference of -0.0311 and the traffic comparison a `mobility_intent` difference of -0.0036. These numbers are reported only to show that the reducer retains metric direction and magnitude. A single LLM-mediated trajectory cannot tell whether the named policy caused the difference, whether the relevant structural mechanism was connected, or whether the sign would persist under a new seed. The case supports a workflow claim—GAWorld can preserve paired trees and compute declared contrasts—rather than a claim about housing, health, training, or transport policy.

12.3 Case 3: Information metrics make inactivity visible

The misinformation artifact contains three five-agent arms. The control and treatment B each cover two simulated days, whereas treatment A covers one. Across every recorded day and arm, average misinformation risk is 0.0, and the reported peak and final risks are also 0.0. Mean cross-viewpoint-exposure is 0.06656 for control, 0.06707 for

treatment A, and 0.06635 for treatment B. The unequal horizons rule out a balanced comparison, and differences at the fourth decimal place are descriptive properties of these files, not estimated intervention effects.

That negative result is nevertheless a useful system diagnostic. The archive records treatment names, horizons, agent counts, daily risk series, and an exposure aggregate together. It does not prove that misinformation propagated, that risk detection was sensitive, or that the interventions changed exposure. The separate polarization artifact reinforces the boundary: its two arms contain 109 and 85 records, respectively, and thus cannot be read as a balanced longitudinal contrast without reconstructing the missingness. v2's value here is inspectability—inactive fields and unequal coverage remain visible rather than being silently converted into a success score.

12.4 Case 4: Economy–wellbeing traces provide dense state, not external validity

The macro-economy archive is the most complete artifact for demonstrating multivariate trace production. Its hashed long-format CSV contains 11,600 rows for five agents, 116 unique recorded steps per agent, and 20 metrics per step ($5 \times 116 \times 20$). The associated configuration describes a three-day, seed-42 run. Direct aggregation of the CSV gives 580 values for each metric across agents and steps. Mean emotion, stress, and economic security are 0.4896, 0.5490, and 0.6270, respectively; their observed ranges are $[0.0233, 0.9310]$, $[0.0907, 0.9898]$, and $[0.4455, 0.8298]$.

The raw row count matters because the accompanying narrative report describes "580 timesteps per agent." In the CSV, 580 is instead the number of agent–step values for one metric (5×116); each agent has 116 distinct step identifiers. We therefore use the CSV cardinalities rather than the report's timestep wording. This discrepancy illustrates the role of an artifact ledger: even a plausible derived report must be checked against its hashed source.

The trace can support new descriptive analyses without a new simulation. For example, partitioning the 116 step identifiers into consecutive ranges 0–38, 39–76, and 77–115 yields emotion means of 0.6365, 0.5020, and 0.3305, while stress means are 0.4122, 0.5428, and 0.6916. These bins are equal-index summaries, not validated "days," and the trajectories remain generated state variables from one seed. They show that GAWorld records joinable, time-indexed state across agents and constructs; they do not show human wellbeing dynamics, economic realism, or an effect of any real-world shock. v2's price-level multiplier is enabled in this archive; the wage index is disabled, and the legend records that omission.

12.5 Case 5: Multi-tenant isolation mechanics

A two-world deployment (one *private*, one *open*) was used to audit the K6 migration. The *private* world is owned by `alice` and is invisible to `bob`; the *open* world is owned by `alice` and lets `bob` join as a player. Both worlds run simultaneously; their `output/` trees are disjoint; their Recorder streams carry distinct `_node` tags when the hub is also a node. The artifact ledger shows that `alice`'s mutation in the *private* world did not affect `bob`'s *open* world and that `bob`'s player intervention in the *open* world was routed to the right node within the poll interval. The audit does not establish a multi-tenant throughput claim: the two-world deployment is a fixture for the isolation mechanics, and a real classroom deployment is needed to exercise the FIFO queue, the per-tenant quota, and the static-file guard. The ledger records this explicitly.

12.6 Case 6: Distributed playable city mechanics

A two-node + hub deployment was used to audit the K7 migration. The hub ran the dashboard, the account store, and the residents assigned to it; one external node ran the remaining residents with the node's own model key. A player (`carol`) joined the *open* world through the browser, claimed a resident on the external node, and performed three interventions over a 30-second window. The artifact ledger shows that each intervention was routed to the right node within the poll interval, that the resident's next perception included the player's action, and that the Recorder merged both nodes' rows into the world's stream with the correct `_node` tags. The deployment also showed that the simulation continued on both the hub and the node when the node's network connection was cut for 30 seconds, and that records were back-filled on reconnection. The audit does not establish a throughput, scale, or fault-tolerance claim: the deployment is a fixture for the protocol mechanics, and a larger deployment is needed to exercise the clock-synchronization median, the per-node outbox, and the per-node audit table.

12.7 Case 7: Track B game archives

The Track B scorecard reports a small number of per-game archives. The *rumor spread* archive contains six runs of a five-resident city with a fixed seed; the *referendum* archive contains four runs of a five-resident city with a fixed seed; the *disaster mode* archive contains three runs of a five-resident city with a fixed seed. In all three cases, fewer than five runs are available for the *post-refutation decline* law, the *conformity under self-interested bill* law, and the *solidarity > panic* law; the scorecard therefore marks the laws as *withheld* and not as *supported* or *reversed*. This is the deliberate Track B behavior: the law gates on a per-game sample size, and the gate is documented.

12.8 Failures are part of the capability record

Two additional archives prevent the coverage matrix from becoming a showcase of successful-looking outputs. The network-evolution report describes five nodes and two initialization edges over a nominal 14-day horizon, but records zero agent–agent interactions on each of the 14 days; the edges were inferred from initial co-location. It cannot support a homophily or emergent-network claim. The emotion-contagion comparison contains four arm entries, all of which report that their state-history CSV is missing. It contains no contagion outcome. Finally, the GAWorld-Bench scorecard matches its built-in synthetic fixture on four contrasts, placebo, and determinism. It verifies that parsers, checks, and report generation execute; it is not a result from GAWorld agents. Reporting these cases alongside the more complete trace is essential to the scientific use of an inspectable system.

13 Validation Boundaries

An artificial society is not validated by the realism of a transcript, by one aggregate match, or by successful execution. GAWorld v2 separates five claim levels: individual consistency, macro correspondence, emergent dynamics, counterfactual response, and reproducibility. Evidence at one level does not compensate for absence at another. This layered view follows established agent-based-modeling practice in distinguishing implementation verification, empirical validation, and the intended domain of use (Grimm et al., 2006; Windrum et al., 2007; Fagiolo et al., 2007), and it now extends the v1 system paper's framework with a sixth column for the participant-evaluation claim level and an explicit evidence-class column for the per-claim ledger.

13.1 Individual consistency

v2 implements the prerequisites for a consistency study: named persistent agents, structured state, episodic files, recall, reflection, longitudinal logs, and a personality plugin that supplies Big-Five proxies for the interruption probability. These mechanisms allow within-agent questions such as whether a stated preference survives intervening events or whether an interview answer is traceable to stored experience. The current memory case does not answer those questions comparatively because three of four arms lack phase 2 and the only complete arm contains one agent. Generated behavioral coherence is also not equivalent to correspondence with a real person. A stronger evaluation would repeat standardized probes before and after controlled memory perturbations, separate temporal persistence from prompt sensitivity, and compare against held-out observations where consent and provenance permit.

13.2 Macro correspondence

Aggregating simulated variables is an implemented capability; calling the aggregate realistic requires a separate calibration argument. The audited economy trace has five agents and one seed, while the benchmark scorecard is a synthetic diagnostic fixture. Neither represents a target population or supports out-of-sample prediction. v2 closes a v1 gap: the price level now actually multiplies real expenditure, and the endogenous inflation rule is a documented driver of the price level. Future macro validation should document the sampling frame, map each simulated construct to an external measurement, reserve targets for testing rather than calibration, and report uncertainty across seeds and plausible parameterizations. Agreement on a few marginal means would remain insufficient if joint distributions, temporal patterns, or subgroup errors failed.

13.3 Emergence and mechanism

An emergent claim requires a pattern that is not merely written into a prompt or initial condition and that recurs under controlled variation. The present archive does not meet that standard: the network study records no interactions, the emotion study has no state files, and the polarization arms have unequal record counts. The existence of social-network, exposure, affect, indoor-room, organization, and governance fields therefore establishes instrumentation, not a social law. Appropriate tests would vary population size and topology, ablate candidate mechanisms, quantify between-seed variability, and trace a macro pattern back to micro-level events. Where a field remains identically zero, the first question is whether the mechanism and detector were active, not why society produced no effect.

13.4 Counterfactual validity

The matched-run interface and the parallel-worlds engine are useful scaffolds for counterfactual experiments, but the archived policy differences are still single contrasts produced by an LLM-mediated dynamical system. A common seed does not guarantee common latent randomness after prompts diverge, and a policy description does not guarantee that an economic or mobility mechanism changed. v2's deterministic counterfactual layer (paired ATE, sign-flip p, BH q, pre-event balance, placebo noise floor, heterogeneity, dose–response, replication, and a non-overturning model interpretation) is an improvement over v1's matched-run reducer, but it does not by itself turn a single LLM-mediated trajectory into a real-world effect. Before interpreting a difference, researchers should verify the intervention's path from event to structural state, replicate both arms over seeds, pre-register target metrics, and include placebo events or negative-control outcomes (Lipsitch et al., 2010). External policy conclusions additionally require a defensible mapping to the population and institutions of interest. GAWorld v2 currently supports constructing and auditing such a study, not replacing it.

13.5 Reproducibility as a non-compensatory gate

The artifact hashes used in this paper bind each numerical statement to an audited byte sequence. v2 extends the v1 ledger with three fields: `world_id`, `node_id`, and `per_arm_status`. These are the minimum additional information needed to reconstruct a run under the v2 deployment model. They do not bind the archive to a clean released runtime: the audit began from a dirty working tree, and hosted providers may change behavior independently of code, seed, or sampling parameters. Missing state files and interrupted arms further show that successful completion must be recorded per treatment. Consequently, a result that cannot be reconstructed from code, configuration, inputs, model identity, world identifier, node identifier, per-arm completion status, and complete outputs should not be promoted because it looks plausible or matches an external statistic.

For future releases, the minimum reproducibility unit should contain the exact code revision plus a patch or clean tag, environment lockfile, initial profiles and maps, provider/model identifiers, prompt and sampling settings, random seeds, per-arm status, logs, raw and derived artifacts, the world identifier, the node identifier (if any), the player intervention trail (if any), and a hash manifest. Hosted-model studies should estimate replay variability rather than promise bitwise determinism. The diagnostic benchmark can then serve as a regression gate: fixtures test parsers and scoring logic, while separately labeled simulation runs test whether scientific evidence has actually been produced.

13.6 Participant evaluation

The Agent Playground and the Agent Arena produce a third class of evidence: per-game or per-question response archives from named residents. The Track B scorecard combines several games under pre-registered laws. Participant evaluation is a distinct claim level because it concerns whether the residents' responses under a fixed scenario follow a rule, not whether the residents approximate a real population. v2 therefore requires that a Track B law gate on a pre-registered minimum sample size; laws that fail the gate are reported as *withheld*, not as *reversed* or *supported*. The Arena is a per-resident benchmark and does not produce a population claim; its archives are reusable but the researcher is responsible for the population inference.

13.7 Intended use

At its present evidence level, GAWorld is best used for mechanism prototyping, trace inspection, hypothesis generation, controlled software experiments, the design of more rigorous artificial-society studies, and the teaching

of agent-based social simulation. The archive does not support deploying GAWorld as a substitute population, forecasting real social outcomes, ranking policies, or making decisions about identifiable people. Those are not merely stronger wordings of the same result; they are different claims requiring external data, identification, uncertainty quantification, and governance that the current artifacts do not provide.

14 Limitations, Ethics, and Roadmap

GAWorld v2 makes a broad set of mechanisms available in one runtime, but breadth does not imply scientific validity or engineering completeness. We separate limitations of the current artifact archive, limitations of the system implementation, and limitations inherent in representing social life through LLM agents. The resulting roadmap likewise separates engineering work that improves the instrument from research work needed to justify substantive claims.

14.1 Evidence and reproducibility limitations

The archived experiments are incomplete and heterogeneous. Several contain a single seed or one agent, treatment horizons differ, some mechanisms produced identically zero fields, and some intended arms lack outcome files entirely. v2's addition of the `warnings.json` file makes the failure modes visible at the artifact level, but it cannot recover missing treatments or provide statistical power after the fact. The existing cases therefore illustrate trace production, comparison mechanics, failure visibility, multi-tenant isolation, hub-node routing, and the per-game probe library, rather than stable behavioral effects. Claims about individual consistency, macro correspondence, emergence, or counterfactual response require purpose-built evaluation beyond the archive summarized here.

Reproducibility is also limited by model and provider dependence. A seed does not freeze a hosted model, its serving stack, safety filters, or routing behavior. Provider fallback may change the model used within a run; library, prompt, or parser versions can change subsequent state; and concurrency can alter call order. In a multi-node deployment, the network round-trip and the per-node queue introduce an additional source of timing variation. Local models improve control but do not eliminate hardware, runtime, and numerical variation. Consequently, meaningful replication requires more than source code and a seed: it requires resolved configuration, prompt and sampling settings, provider and model identifiers, dependency versions, active plugins, warnings, per-arm completion status, world and node identifiers, player intervention trails, and raw artifacts. Hosted-model studies should measure replay variability rather than promise bitwise determinism.

Runtime and monetary cost constrain the studies that can be performed. A persistent society may invoke generation, embedding, retrieval, reflection, environment, consolidation, organization, governance, and information-source paths repeatedly across agents and ticks. v2's addition of an in-memory *job store* for game sessions reduces the per-game overhead, and the per-tenant quota lets an admin throttle a misbehaving tenant, but neither change reduces the model-call cost of a meaningful study. Cost pressure can also encourage small populations and short horizons that are inadequate for emergence or uncertainty analysis. Future evaluations should report call counts, latency, failures, model routes, and estimated cost alongside simulated scale and horizon.

14.2 Architecture and data limitations

The K1–K5 microkernel migration is complete for its defined scope; K6, K7, and K8 are complete for their respective migrations (multi-tenant world resolution, hub-and-node distribution, parallel-worlds execution). Compatibility paths nevertheless remain: memory, environment, and social logic are not wholly plugin-owned, the Controller does not mediate every action, and the Recorder is not the sole provenance stream. Shared dictionaries and the context escape hatch weaken schema enforcement and ownership boundaries. The in-process distributed relay is best-effort and does not provide transactional consistency; the parallel-worlds engine is sequential at the world level and does not run worlds in parallel. Third-party plugins run in process and should not be treated as sandboxed. A configured extension point therefore does not guarantee mechanism isolation, security, or semantic equivalence under replacement.

The artifact layer has corresponding schema plurality. Agent logs, long-form state CSVs, environment timelines, memory stores, indoor spatial trees, organization ledgers, governance ledgers, world-rooted Recorder streams, and parallel-worlds result files use different schemas and sometimes different notions of step or time. v2 adds a hash manifest (`manifest.json`) to each world root that binds the world's active configuration, the active plugin list, the seed, and a content-addressable list of the world's artifact files; this manifest is the v2 minimum for "this is the

artifact I am looking at". Cross-subsystem analysis still requires explicit joins and can silently misread cardinality or missingness. The discrepancy identified in the economy case between report wording and raw CSV cardinality illustrates this risk. Until a versioned provenance schema and manifest bind clocks, entities, mechanisms, and derivations, replay and analysis must remain artifact-aware.

Input profiles are another boundary. They may be sparse, stylized, synthetic, distilled from a real person, or assembled from data whose completeness and provenance vary. v2's persona-distillation module adds a sourced-portrait intermediate; rich generated backstories do not repair missing observations, they can turn uncertainty into plausible detail. Profile coverage, source, consent, transformation, and missingness should be documented, and downstream readers should distinguish supplied attributes from model-generated elaboration.

14.3 Ethical boundaries and affected communities

Synthetic agents can create an illusion of access to perspectives that were never actually observed. Language models may reproduce stereotypes, suppress within-group heterogeneity, and make marginalized identities appear more uniform or legible than they are (Wang et al., 2025). Conditioning a model on demographic labels or a short profile does not confer lived experience, community membership, consent, or authority. Even when aggregate outputs resemble survey responses in one setting, LLM agents should not be treated as drop-in replacements for human participants (Dillion et al., 2023).

GAWorld therefore must not substitute for engagement with people affected by a research question or policy. It cannot determine community priorities, adjudicate contested values, estimate acceptable harms, or legitimate a decision about housing, health, employment, policing, mobility, or information governance. For consequential applications, affected communities should help define the question, categories, outcomes, unacceptable uses, and release conditions; domain experts and empirical data are needed to evaluate whether the modeled mechanisms correspond to the institutions at issue. Simulation may support deliberation or identify assumptions to test, but it cannot replace participation, due process, or accountable public decision-making.

Profiles, memories, interviews, and generated social ties can also expose or infer sensitive information. v2's persona-distillation module separates *distill* from *deploy*: the distilled portrait is held under `output/personas/<slug>/` and only an explicit deploy step writes the seed CSV that a researcher can publish. The deploy step should be reviewed, and the underlying source material should be subject to the same consent and transformation rules as the original data. Dashboard, relay, and environment services are local research tooling rather than production-secure deployments; exposing them requires authentication, encryption, authorization, audit, and abuse controls. Generated claims about real people should not be presented as observations, and simulated agents should not be used to rank, target, or make decisions about identifiable individuals.

14.4 Engineering roadmap

The engineering roadmap is to make current boundaries enforceable and current artifacts easier to reconstruct. Priorities are: (1) extend typed Controller validation beyond movement to actions and broadcasts where governance is required; (2) migrate remaining domain mechanisms (environment evolution, social-network mechanics, memory consolidation) incrementally behind declared dependencies without changing their semantics silently; (3) replace shared dictionary conventions with versioned schemas and explicit state ownership; (4) unify trace identifiers and clock semantics through a manifest-backed provenance layer while retaining domain-specific payloads; (5) capture model, provider, prompt, fallback, dependency, plugin, world, node, and player-intervention resolution in every run; (6) add conformance, failure-injection, migration, security, and replay tests; and (7) introduce a conservative-synchronization option for distributed deployments that need a tighter clock alignment than the current best-effort protocol. These changes would improve composability and auditability. They would not, by themselves, validate an artificial society.

14.5 Research roadmap

The research roadmap begins only after the execution unit is reproducible. For individual consistency, standardized repeated probes and controlled memory perturbations should be evaluated against held-out criteria where consent and provenance permit. For macro correspondence, researchers should define a sampling frame, map simulated constructs to external measures, separate calibration from testing, and quantify subgroup error and uncertainty. Claims of emergence require multiple seeds, scales, topologies, and mechanism ablations, with macro patterns traced to micro events. Counterfactual studies require verified intervention paths, replicated arms, pre-specified

outcomes, placebos or negative controls, and an explicit account of transportability. Provider and prompt sensitivity should be treated as experimental factors, not hidden implementation noise. Participant-evaluation studies should pre-register the minimum sample size, document the scoring rule, and report the gate result rather than a forced yes/no.

Finally, future studies should evaluate not only whether a simulated pattern is reproducible, but also whether the representation and intended use are appropriate. Participatory problem formulation, impact assessment, and documentation of excluded perspectives belong beside technical validation. GAWorld's near-term role is therefore to make artificial-society hypotheses, mechanisms, traces, and failures easier to construct and inspect—not to turn generated populations into unearned evidence about society.

15 Conclusion

We have presented GAWorld v2, a research stack for building, auditing, and revising LLM-based artificial societies. v2 extends the v1 platform along five dimensions: a fourteen-plugin microkernel with a closed K1–K8 migration; multi-tenant, per-world, role-aware deployment; hub-and-node distributed execution and playable cities; a Research Workbench that compiles a paper or research idea into a pre-registered protocol and a deterministic counterfactual verdict; and an Agent Playground of fourteen game-based probes and an Agent Arena. v2 also closes a long-standing gap between documented and implemented behavior in the price-level module, adds an indoor spatial tree, and persists organizations and governance as a separate ledger from the per-step metrics.

The contribution of v2 is not a new cognitive architecture, a new counterfactual estimator, or a new social theory. It is a configurable instrument whose plugins, hooks, and surfaces make the construction and audit of an artificial-society experiment a tractable engineering task. Every reported number in this paper is bound to a hash, an evidence class, and the producing code path; every failed arm is reported as a failure, not as a partial finding. The five claim levels that v1 introduced are extended with a sixth (participant evaluation) and remain the boundary against which substantive social claims are held. GAWorld v2 is offered as a substrate for the research community, not as a substitute for it: the construction, the audit, and the revision of artificial-society hypotheses remain the work of the researcher, and the platform is only as good as the questions it is asked to make auditable.

Three directions are particularly open. First, a more comprehensive *individual-consistency* evaluation that uses repeated standardized probes before and after controlled memory perturbations, with a pre-registered analysis. Second, a more comprehensive *macro-correspondence* evaluation that maps a small number of simulated constructs to external measures, with a separate calibration sample and a separate testing sample. Third, a more comprehensive *participant-evaluation* evaluation that uses the Track B scorecard to test the three pre-registered laws under sufficient sample sizes, and adds two more laws (rumor verification; referendum partisanship) under the same gate. These are the research steps that v2 makes tractable and that v3, if it is written, will report on.

The platform's source code, the artifact ledger, the per-arm completion audits, the Track A and Track B scorecards, the Research Workbench pre-registered protocols, and the parallel-worlds counterfactual results are versioned together; a reader who wants to inspect a claim can follow the hash, the evidence class, and the world and node identifiers back to the producing code path. This is the level of audit that we believe the field of LLM-based social simulation should aspire to, and the level at which GAWorld v2 is offered.

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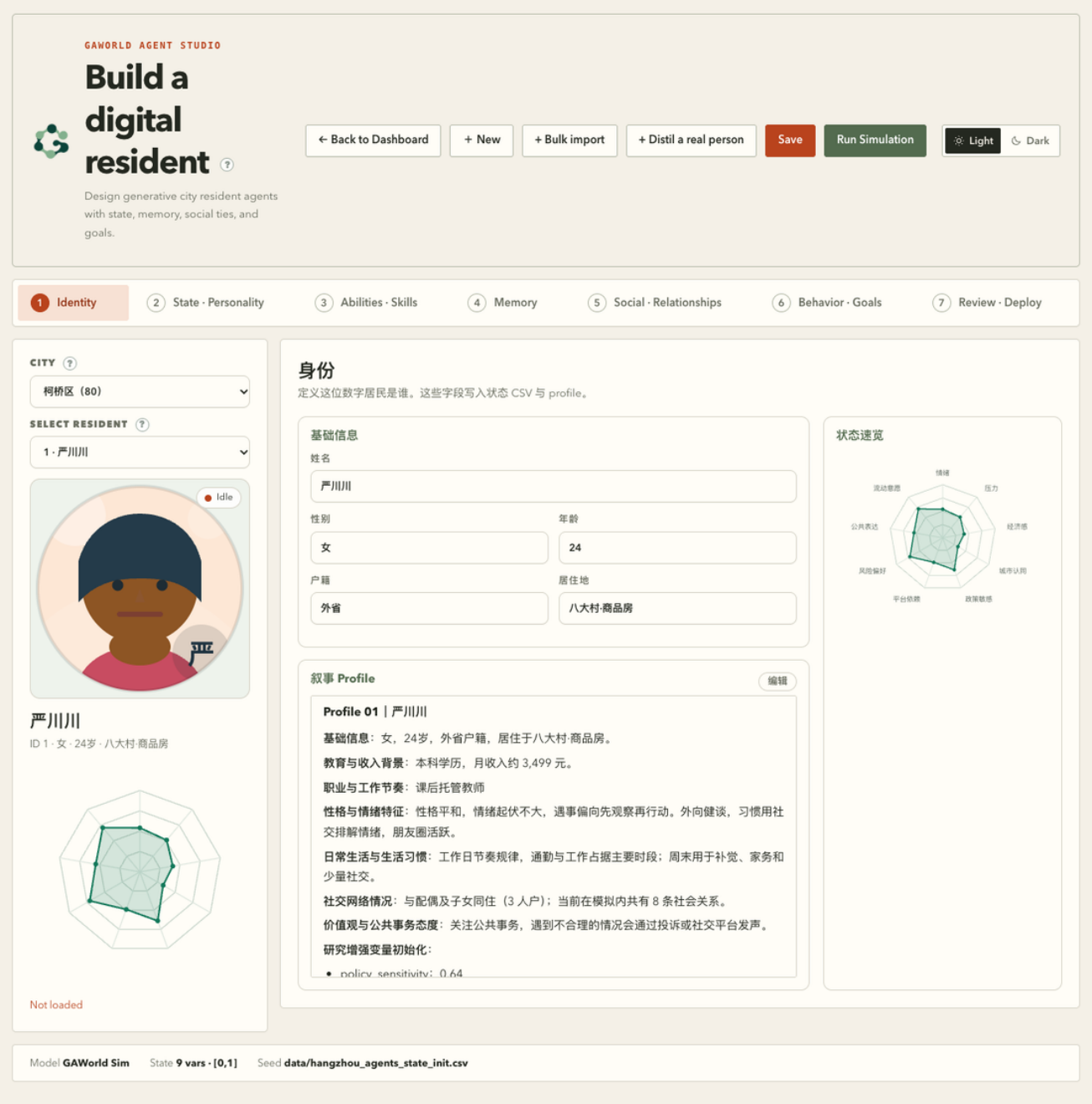


Figure 3: The Agent Studio. The seven-step tab at the top (*Identity, State & Personality, Abilities & Skills, Memory, Social & Relationships, Behavior & Goals, Review & Deploy*) maps one to one onto the seven sections of an Agent Studio session. The left column selects the city and the resident; the right radar plots the normalized state across nine axes (activity, public-expression, pressure, economic sense, city recognition, policy sensitivity, risk preference, platform dependence, mobility intent). The *Profile* pane is the sourced-portrait view; the persona distillation module can write a similar portrait from a name or a URL through the *Distill a real person* button on the top right.

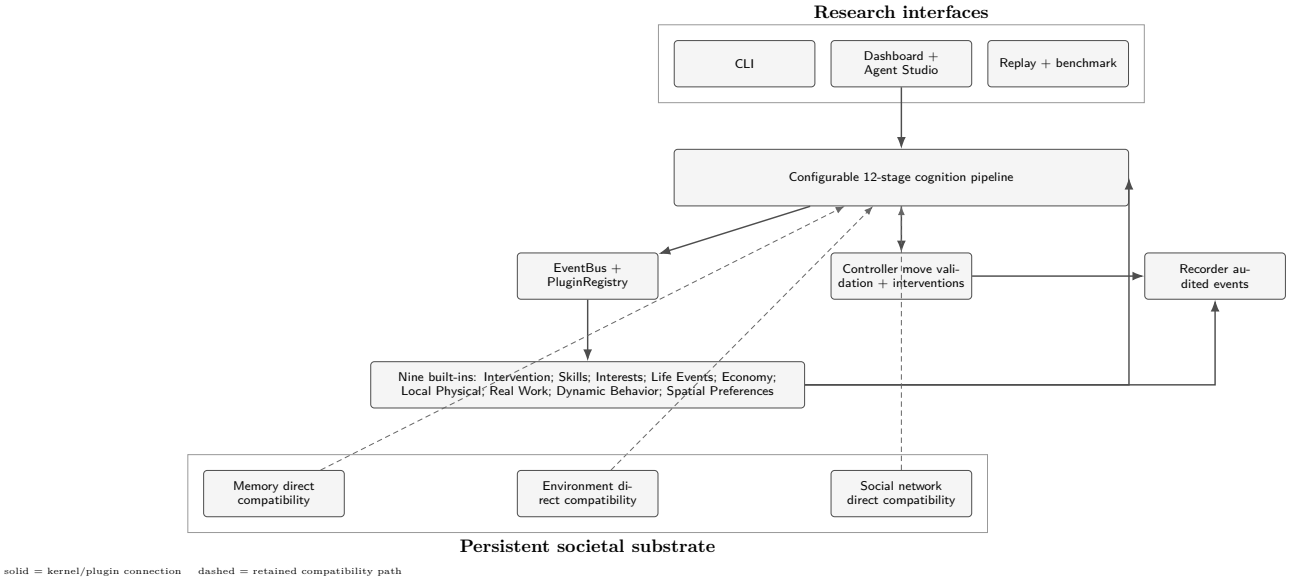


Figure 4: Audited v2 execution architecture. The kernel (Clock, EventBus, PluginRegistry, Controller, Recorder, SimContext) serves the cognition pipeline and the per-tenant world resolver. Fourteen built-ins contribute through the registry, bus, pipeline filters, or lifecycle events. Movement uses the Controller validation gate; Recorder events coexist with established domain-specific outputs. In a multi-tenant deployment, the world resolver is the request-time entry point; in a multi-node deployment, the hub routes player interventions to the owning node.

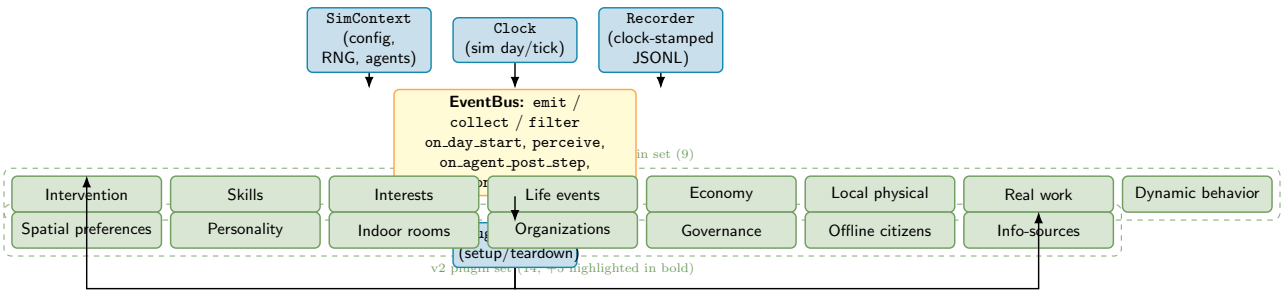
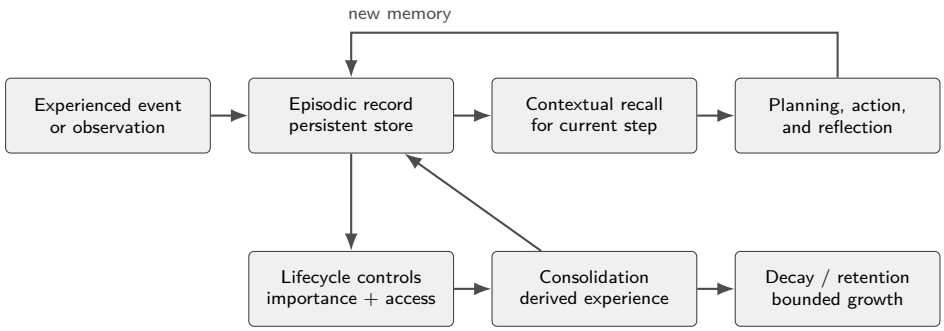


Figure 5: The fourteen v2 built-in plugins and their relationship to the kernel. Plugins may subscribe to lifecycle events, contribute to perception, filter a value, or own namespaced state. The v1 plugin set (nine) is shown above the v2 additions (five) for comparison.



Conceptual lifecycle: persistence enables longitudinal state; retention and consolidation prevent an unbounded raw transcript.

Figure 6: Persistent experience and growth paths. Tick-level episodes feed retrieval and rule-based updates; configured day-boundary processes may consolidate episodes into semantic memories or private skills, classify growth into a phase, drain the life-event queue, and tally governance ballots. Solid storage paths denote implemented persistence, not validated models of human remembering or learning. Dashed paths indicate optional LLM calls that may be disabled in a controlled experiment.

Table 3: Coupling among agent and societal components. Evidence status describes implementation visibility, not empirical validity.

Subsystem	Reads	Writes	Agent-visible consequence	Evidence status
Memory and cognition	Episodes, state, context, retrieved entries	Memories, intentions, habits, relationships	Recalled context, priorities, reflection, later action bias	Implemented; human realism unvalidated
Interests and skills	Profile, episodes, social growth context	Growth profile, private skills	Practice focus and procedural prompt context	Implemented; partial plugin boundary
World and local physics	City graph, time, weather, locations, rooms	Routes, travel state, occupancy snapshots, room occupancy	Feasible destinations, travel costs, crowding, closure cues, room-level co-location	Implemented; direct integration
Environment	Configuration, history, optional LLM output	Events, weather and macro context	Interruptions and perceived exogenous conditions	Implemented; generator-mode dependent
Social network	Profiles, contact history, interactions	Typed ties, closeness, obligation, tiers	Encounter probability and relationship context	Implemented; mechanism validity unassessed
Life events	Persisted event queue, clock	Consumption flag and configured state effects (typed, reversible)	Scheduled exogenous changes to targeted agents	Implemented; committed plugin-integrated event producer
Economy	Activity, balances, calendar, sectors, macro state, price level	Accounts, ledger, budgets, pools, agent state, end-of-month price index	Income, spending constraints, taxes, credit, returns, payments, real-consumption erosion	Implemented; primarily hook-based
Work	Capability cache, activity, growth, market jobs	Queue results, artifacts, settlement and state effects	Produced files, earnings, experience records	Implemented; artifact quality unvalidated
Indoor rooms and capacity	City map, location, activity	Per-room occupancy, capacity enforcement, room-intent binding	Room-resolved perception, capacity-based rejection	Implemented; opt-in
Personality	Profile, optional LLM overlay	Personality proxies	Conditional interruption probability, growth-phase weighting	Implemented; anchor + overlay
Organizations and governance	Membership, role, ruleset, proposal	Updated ruleset, accepted proposal, vote tallies	Rule-conformant behavior, post-vote transitions	Implemented; opt-in
Offline citizens and travel	Off-screen contact roles, duty obligations	Visit channel, trip itinerary, location rewrite	Mobile-agent perception, family duty fulfillment	Implemented; opt-in
Information sources	Channel list, search budget, agent-derived query	Retrieved entries, Moltbook posts, perception field	Exposure heterogeneity, search-induced awareness	Implemented; channel-dependent

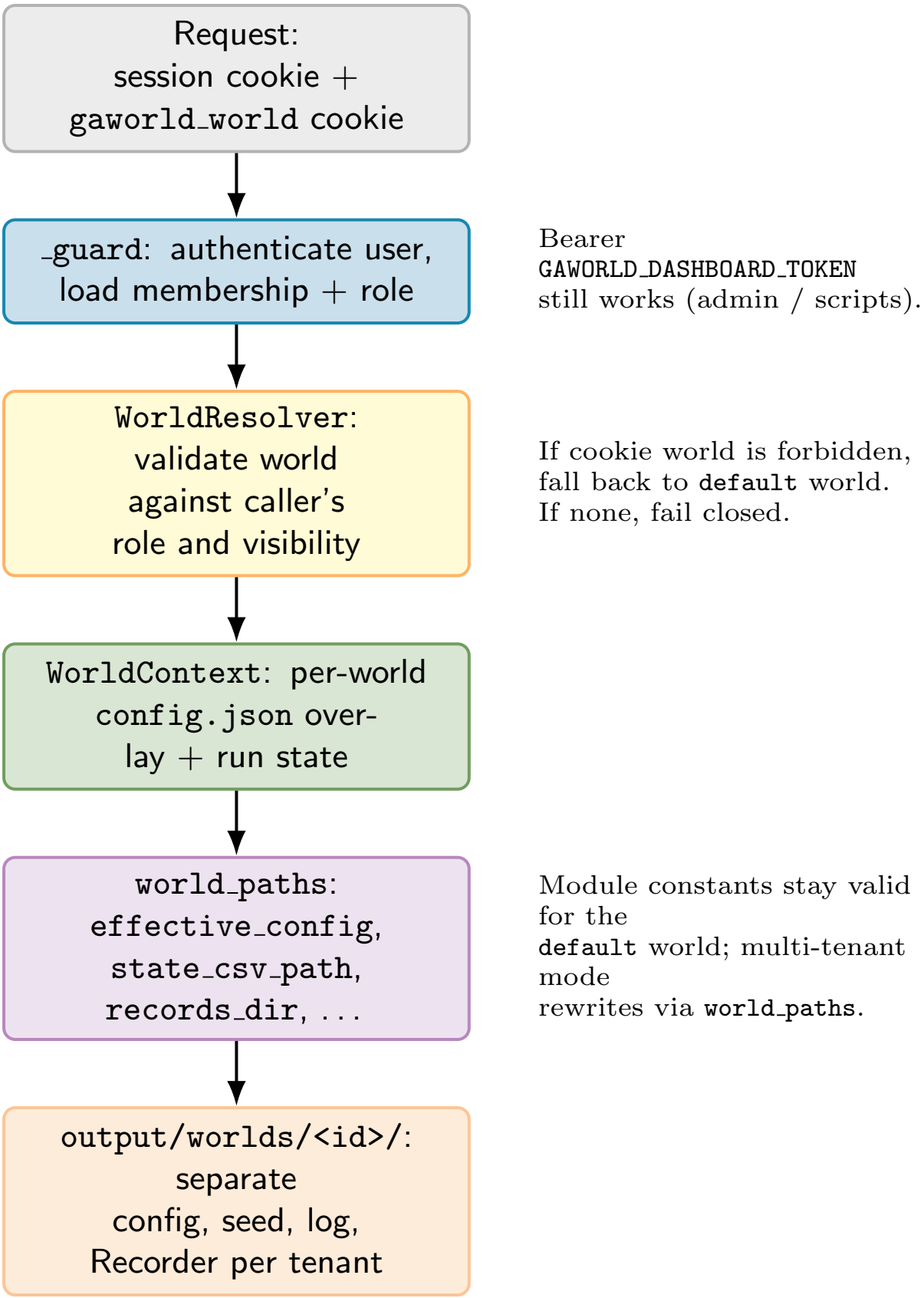
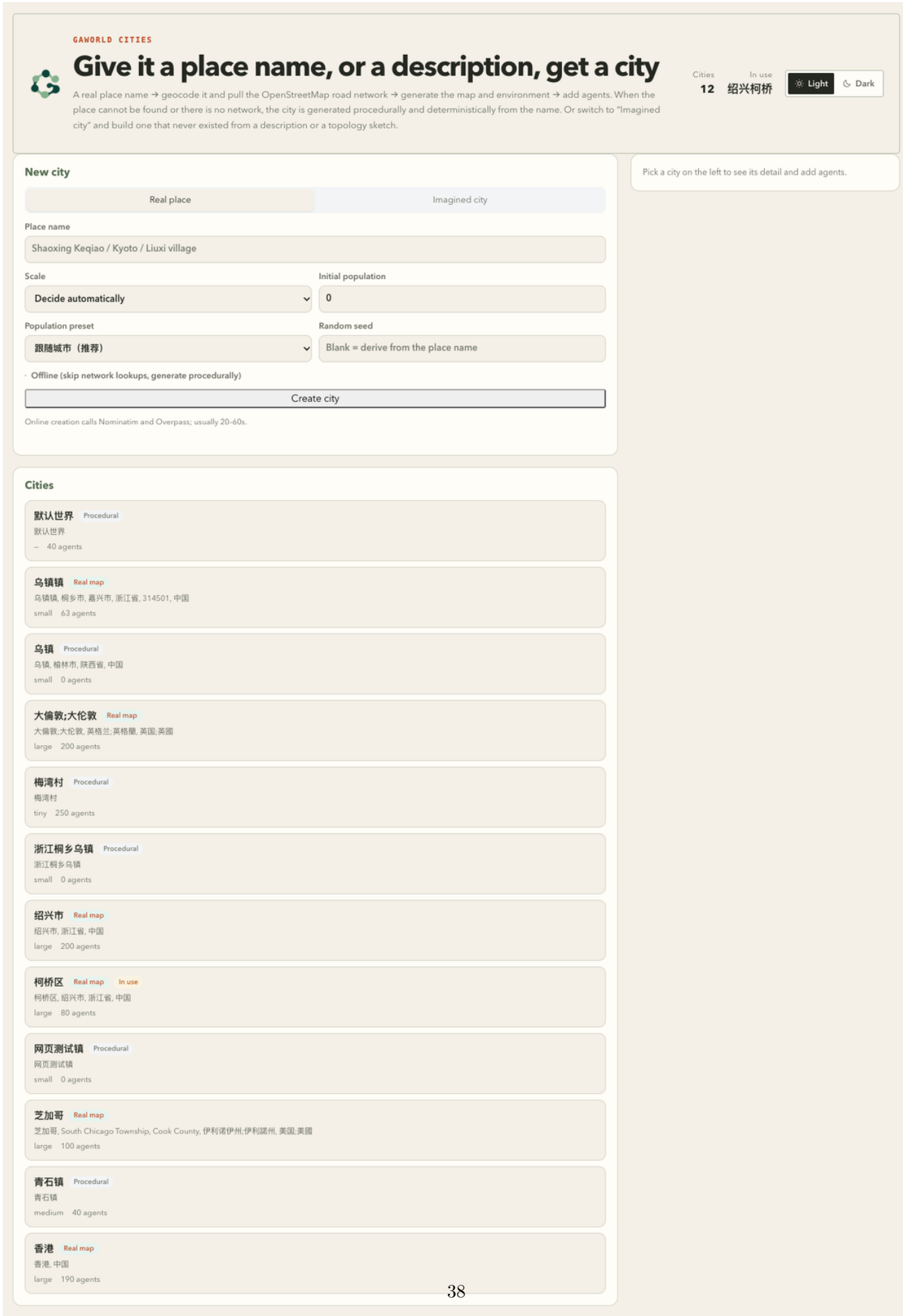


Figure 7: Request resolution in a multi-tenant deployment. A request enters the dashboard server with a session cookie and a `gaworld_world` cookie; `_guard` authenticates the caller, the `WorldResolver` validates the cookie against the caller's role and visibility, and the resulting `WorldContext` drives per-tenant path resolution.


GAWORLD CITIES
Give it a place name, or a description, get a city
A real place name → geocode it and pull the OpenStreetMap road network → generate the map and environment → add agents. When the place cannot be found or there is no network, the city is generated procedurally and deterministically from the name. Or switch to "Imagined city" and build one that never existed from a description or a topology sketch.
Cities
12
In use
绍兴柯桥
Light
Dark
New city
Real place
Imagined city
Place name
Shaoxing Keqiao / Kyoto / Liuxi village
Scale
Decide automatically
Initial population
0
Population preset
跟随城市（推荐）
Random seed
Blank = derive from the place name
Offline (skip network lookups, generate procedurally)
Create city
Online creation calls Nominatim and Overpass; usually 20-60s.
Pick a city on the left to see its detail and add agents.
Cities
默认世界 Procedural
默认世界
– 40 agents
乌镇镇 Real map
乌镇镇, 桐乡市, 嘉兴市, 浙江省, 314501, 中国
small 63 agents
乌镇 Procedural
乌镇, 榆林市, 陕西省, 中国
small 0 agents
大倫敦;大伦敦 Real map
大倫敦;大伦敦, 英格兰;英格蘭, 英国;英國
large 200 agents
梅湾村 Procedural
梅湾村
tiny 250 agents
浙江桐乡乌镇 Procedural
浙江桐乡乌镇
small 0 agents
绍兴市 Real map
绍兴市, 浙江省, 中国
large 200 agents
柯桥区 Real map In use
柯桥区, 绍兴市, 浙江省, 中国
large 80 agents
网页测试镇 Procedural
网页测试镇
small 0 agents
芝加哥 Real map
芝加哥, South Chicago Township, Cook County, 伊利诺伊州;伊利諾州, 美国;美國
large 100 agents
青石镇 Procedural
青石镇
medium 40 agents
香港 Real map
香港, 中国
large 190 agents

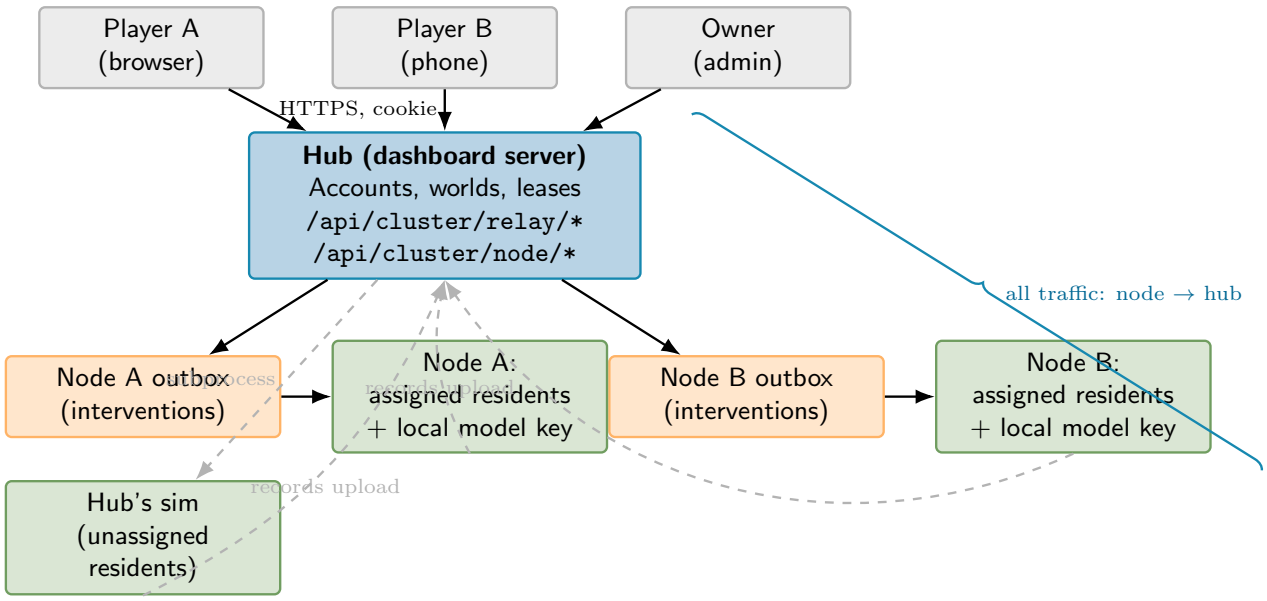


Figure 9: Distributed playable topology. The hub is the dashboard server; it owns accounts, the world metadata, the per-node outbox, the per-world relay, and the per-tenant Recorder merge. Each node runs the simulation for its assigned residents, polls interventions, and uploads Recorder rows. All cross-machine traffic is node-to-hub; the hub never opens inbound ports to a node.

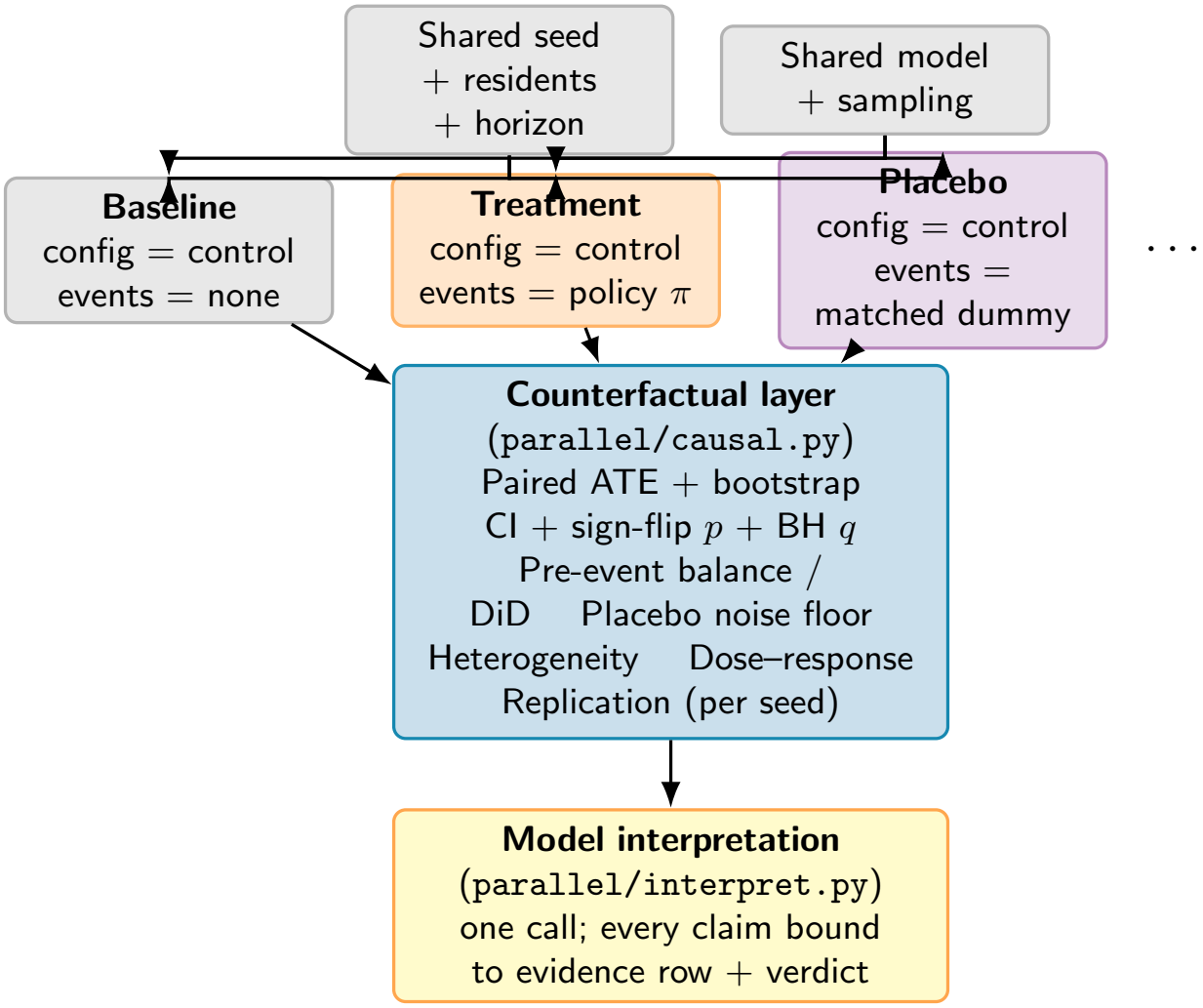


Figure 10: Parallel-worlds engine. Up to eight matched worlds share the same seed, the same resident sample, and the same model. Each world receives a per-world `config` patch and a per-world event list; the counterfactual layer then computes paired ATE, sign-flip p, pre-event balance, placebo noise floor, heterogeneity, dose–response, and replication. An optional one-call model interpretation binds every claim to a specific evidence row.

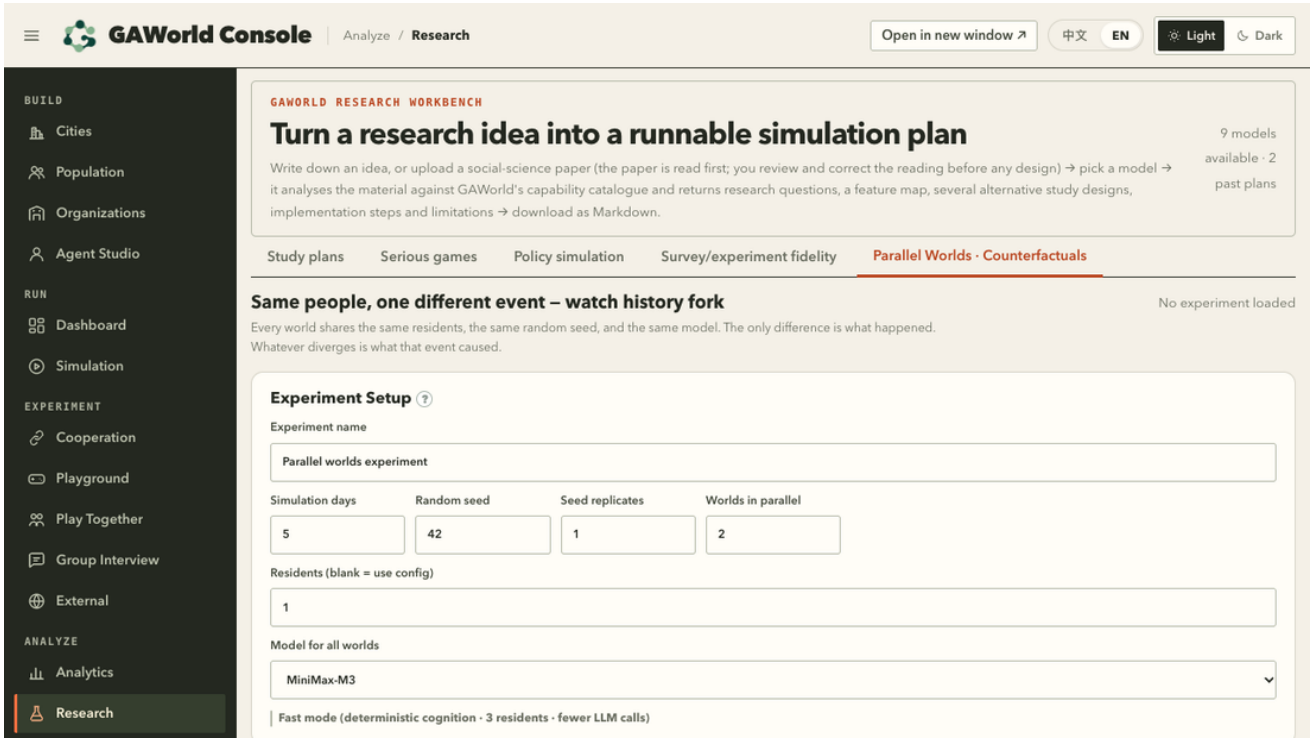


Figure 11: The Parallel Worlds tab. A researcher fills in the experiment name, the simulation days, the random seed, the number of seed replicates (for the per-seed ATE and meta-analytic mean), the number of worlds to run in parallel (up to eight), the resident subset, and the model. The *Fast mode* toggle switches the engine to deterministic cognition with a smaller resident set so that the workflow can be rehearsed cheaply before a real run.

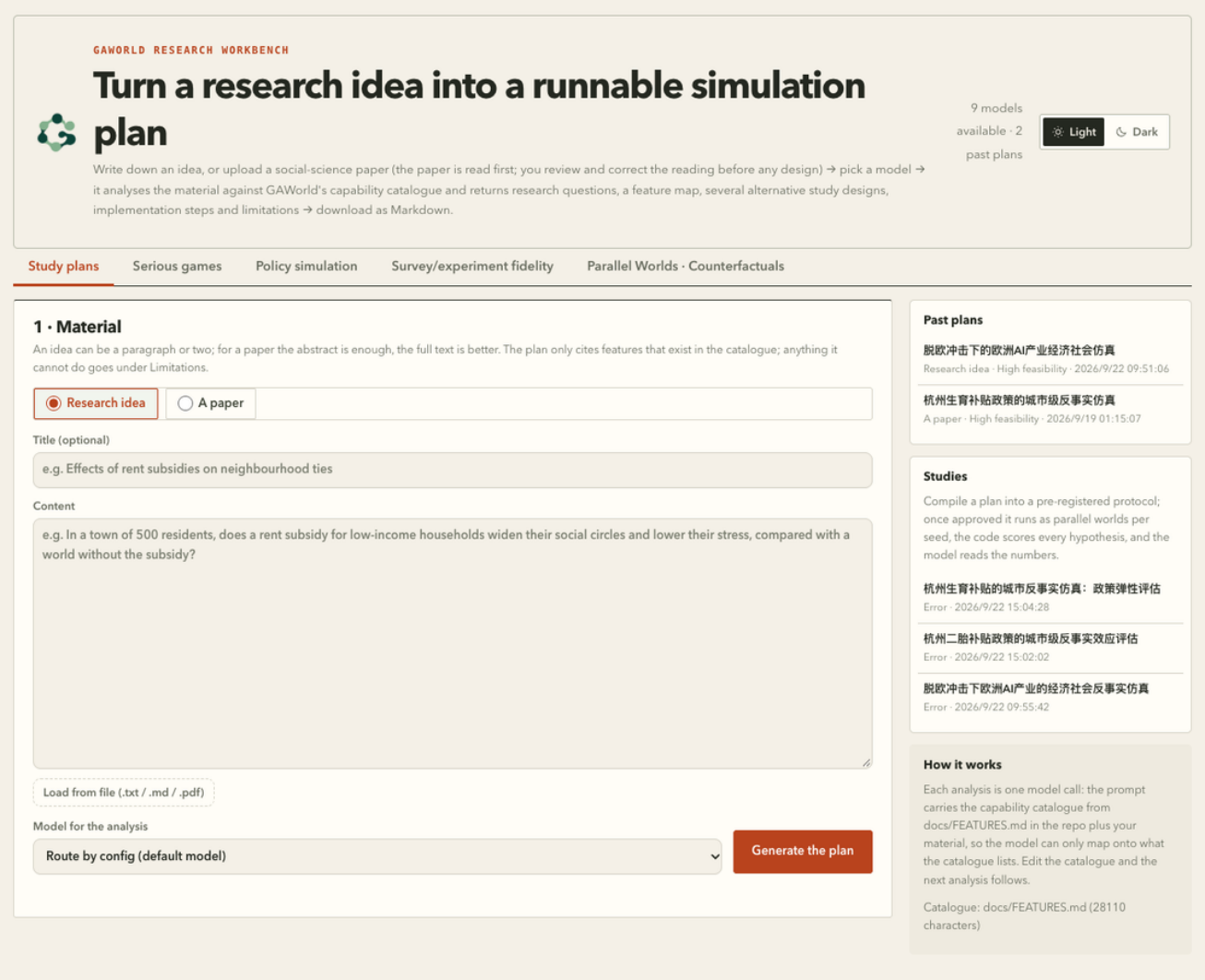


Figure 12: The Research Workbench. A researcher pastes a research idea or uploads a paper, picks a model, and the system returns a plan: a theory, a methods paragraph, key variables, hypotheses, a feature map to the GAWorld surface, two or three experimental designs, and a list of limits. The plan is editable; the model cannot write to a feature that is not in `FEATURES.md`, and the protocol compiler cannot use a metric that is not in the *measurable catalog*. The *Study plans*, *Serious games*, *Policy simulation*, *Survey / experiment fidelity*, and *Parallel Worlds Counterfactuals* tabs reuse the same pane to compile plans, run serious-game sessions, and run paper-fidelity studies.

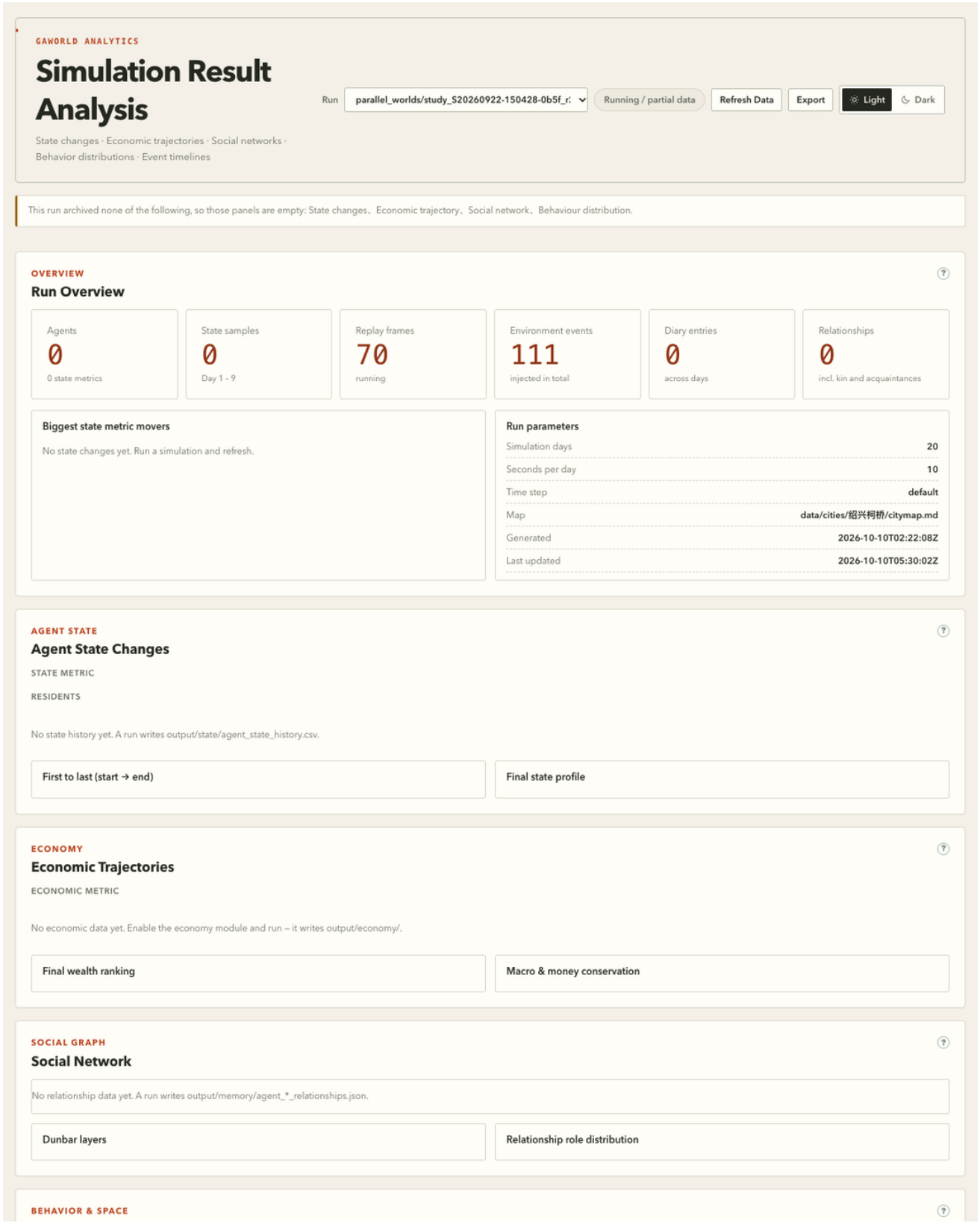


Figure 13: The Analytics pane. After a run, the analytics pane exposes per-run counts (agents, state samples, replay frames, environment events, diary entries, relationships), a per-day event timeline broken down by source (*social / natural / economic / political / life_event / technology*), an event-type distribution, and an impact-tag distribution. When a run is incomplete or a metric is missing, the pane displays *No data yet* in the corresponding block rather than fabricating a value, in line with the evidence-class policy.

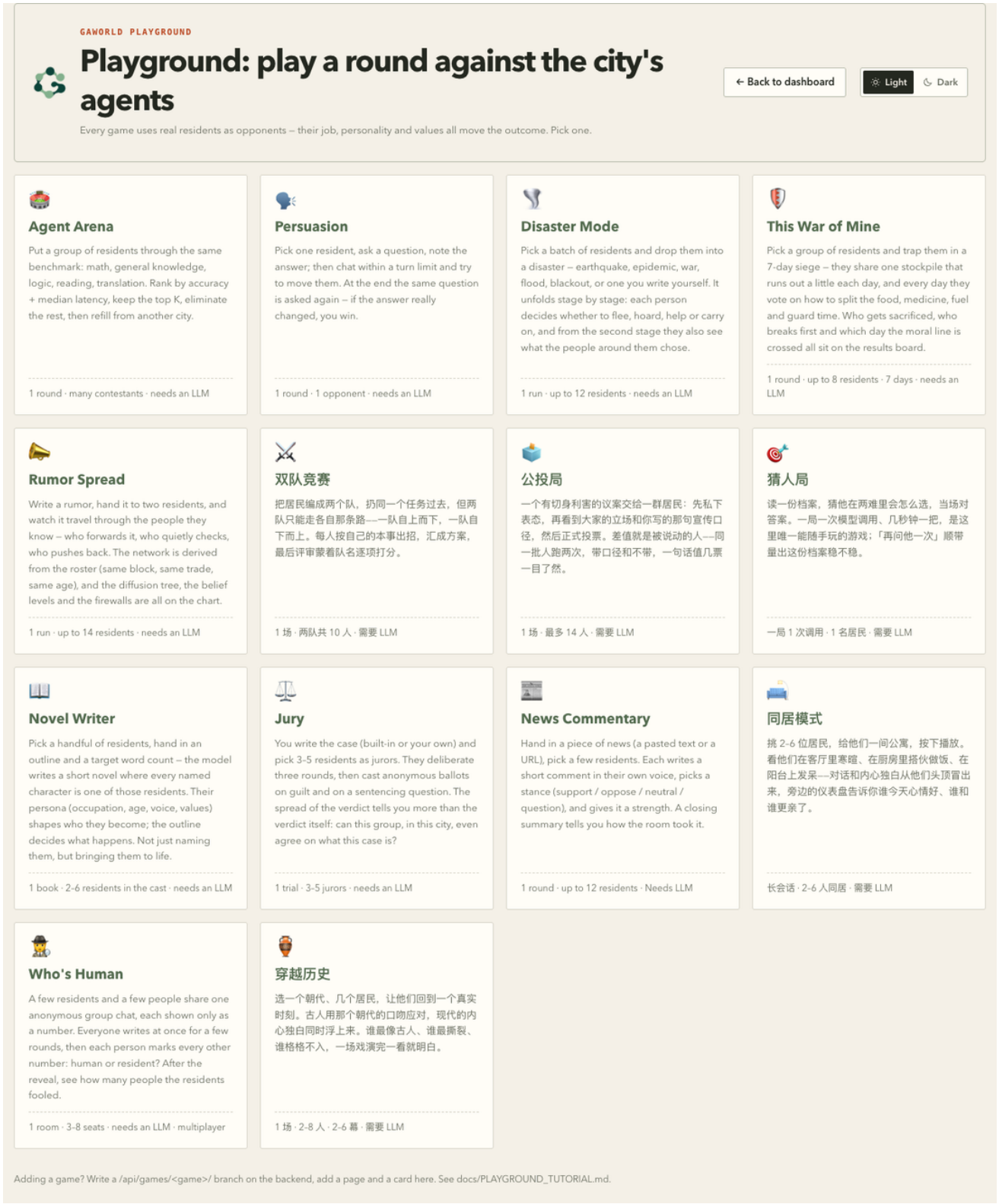


Figure 14: The Agent Playground. Each card is a game-based probe; the fourteen cards in the v2 set cover the four families (one-on-one, one-group-one-event, mission / creative, and long-form / human-in-the-loop). Every card is a deterministic harness over a real resident's response; per-game archives are written to `output/games/<game>/` on completion and feed the Track B scorecard.

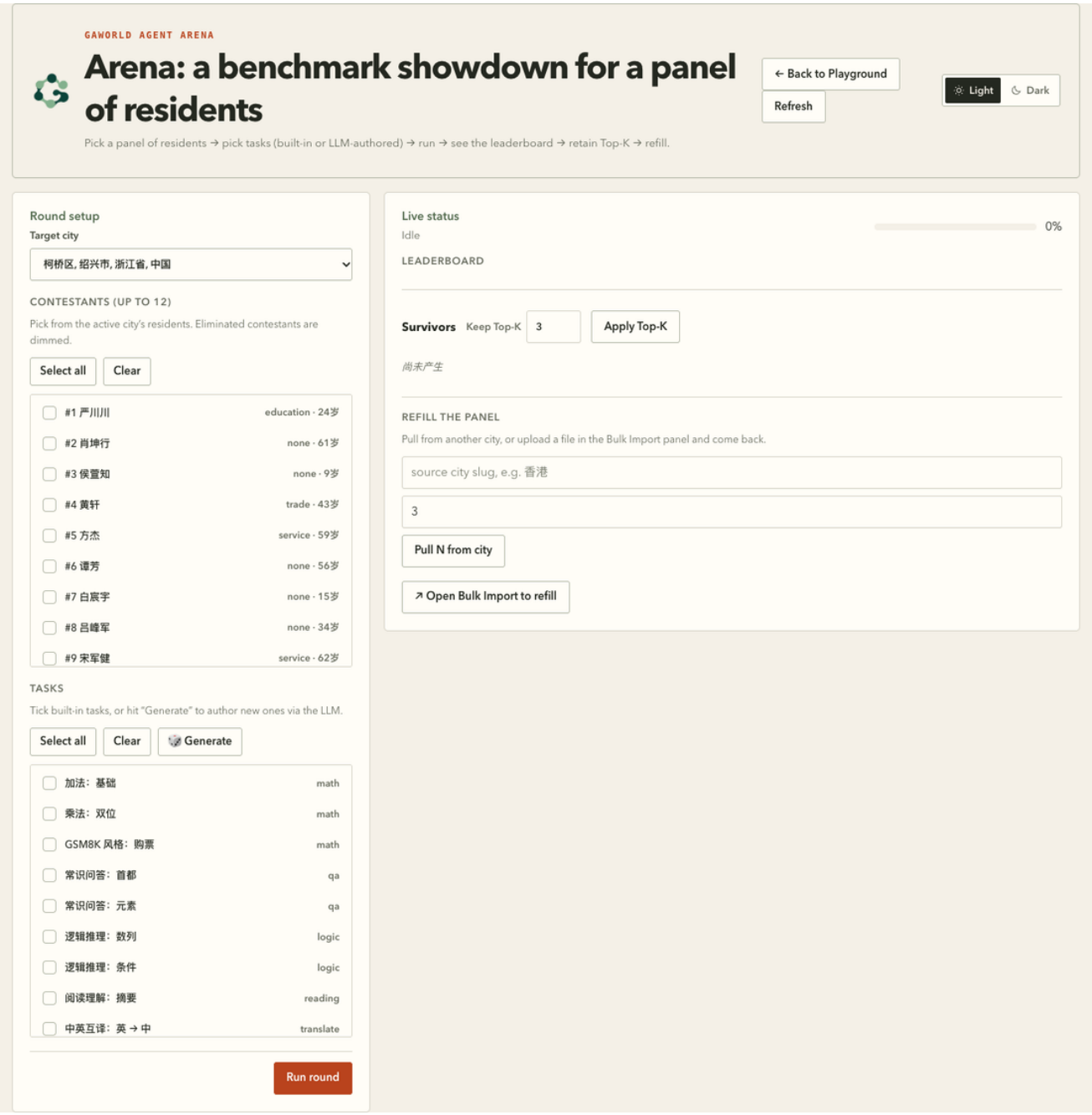


Figure 15: The Agent Arena. A researcher picks a target city, a panel of up to twelve residents, and a question set (a built-in bank of ten questions or an LLM-generated set). After the round, the leaderboard ranks the residents by correctness and median latency. A *Keep Top-K* policy retires the bottom of the panel; the *Refill the panel* controls pull new residents from another city or from a bulk-import file.

Capability area	Archive status	What the current artifacts support
Macro economy + wellbeing	COMPLETE	Multivariate longitudinal trace production
Persistent-memory study	PARTIAL	Cross-phase artifacts; incomplete treatment coverage
Policy comparisons	PARTIAL	Matched event/baseline workflow; single-run evidence
Information interventions	PARTIAL	Exposure and stance metric pipeline; inactive fields disclosed
Network evolution	PARTIAL	Initialization trace; interactions insufficient
Emotion contagion	DESIGNED	Protocol exists; archived outcome evidence incomplete
ABM validation	DESIGNED	Validation workflow exists; completed evidence insufficient
GAWorld-Bench scorecard	FIXTURE	Software-path diagnostics only, not simulation findings

Statuses describe evidence availability, not comparative performance or scientific validity.

Figure 16: Coverage of the audited v2 result archive. The matrix distinguishes inspectable capability evidence from incomplete evaluations and diagnostic fixtures; it is not a leaderboard or a validation score.

Table 4: Status of all experiment families represented in the v2 artifact ledger. Counts and horizons describe the archived files, not a common evaluation protocol.

Area	Status	Archived scope	Permitted use
Macro economy and wellbeing	REAL	$n = 5$ agents, one seed, three-day configuration; 116 recorded steps per agent	Multivariate state-trace production, not an economic or psychological finding
Persistent memory	INCOMPLETE	$n = 1$ agent; four planned 14-day arms; only one completed both seven-day phases	Persistence and cross-phase artifact inspection, not a treatment comparison
Policy comparisons (v1 archive)	REAL	Four single archived event–baseline pairs; Day-3 events; no seed replications	Pairing, isolation, and metric extraction, not policy effects
Information intervention	INCOMPLETE	Three arms, $n = 5$ each; unequal horizons of one or two days	Information-metric schema and missing/inactive-field audit
Polarization	INCOMPLETE	Two arms with 109 versus 85 records	Availability of a comparison artifact; unequal records preclude an effect claim
Network evolution	INCOMPLETE	$n = 5$, 14-day report; zero recorded interactions on every day	Initialization and failure visibility, not emergent-network evidence
Emotion contagion	INCOMPLETE	Four intended arms; all four report a missing state-history file	Protocol coverage only; no outcome result
ABM validation	INCOMPLETE	Validation workflow is present; completed empirical evidence is insufficient	Evaluation design only, not an empirical validation result
GAWorld-Bench Track A	FIXTURE	Built-in synthetic scorecard	Parser and scoring-path diagnostics only
GAWorld-Bench Track B (rumor / referendum / disaster)	REAL, INCOMPLETE	A handful of per-game archives; not enough runs to test the three preregistered laws	Probe library, not population evidence
Parallel-worlds sweeps	DIAGNOSTIC	Cached per-arm artifact trees from a recent transport-restriction sweep	Counterfactual-inference code paths; not policy evidence
Multi-tenant isolation	REAL	Two-world deployment with a single dashboard, two world roots, two Recorder streams	Isolation mechanics, not scale claim
Distributed playable city	REAL	Two-node + hub deployment with one world, residents split between hub and node, player interventions routed across nodes	Hub-node protocol mechanics, not throughput claim
Persona distillation	REAL	One distilled portrait, one deployed resident	Distill-and-deploy workflow, not population claim

Table 5: Claim–evidence matrix for the audited v2 snapshot. "Current support" states the strongest defensible claim, not an aspirational goal. The "Evidence class" column is the class of the strongest available artifact, not a statement that the strongest artifact is also a population claim.

Claim level	Evidence required	Current support	Not established	Class
Individual consistency	Repeated independent elicitations; longitudinal and perturbation tests; external or held-out criteria	Persistent profiles, memories, diaries, interviews, and one completed two-phase arm are inspectable	Human fidelity, stable identity across models/seeds, or a memory-treatment effect	REAL / INCOMPLETE
Macro correspondence	Pre-specified external targets; calibrated population; uncertainty and out-of-sample tests	State/economy pipelines can produce aggregates; benchmark fixtures test analysis code	Population representativeness, calibrated macro fit, forecasting, or real-city prediction	REAL / FIXTURE
Emergent dynamics	Multiple seeds and scales; mechanism ablations; network/temporal diagnostics; robust qualitative patterns	Social and network mechanisms emit inspectable state; failures are archived	Polarization, contagion, homophily, or other stable emergent laws	REAL / INCOMPLETE
Counterfactual response	Replicated randomized arms; verified intervention path; placebo and negative controls; uncertainty	Four paired event/baseline trees (v1) and the parallel-worlds engine (v2) demonstrate comparison mechanics	Causal policy effects, sign stability, transportability, or welfare recommendations	REAL / DIAGNOSTIC
Reproducibility	Immutable code/config/data/model snapshots; deterministic components; repeated reruns and failure accounting	Hashed result files, seeds/configs, per-arm completion, world and node identifiers, the diagnostic harness, and the Track A scorecard	Bitwise replay of hosted LLM calls or complete end-to-end reproduction of every archive	REAL / FIXTURE
Participant evaluation	Fixed scenarios; deterministic scoring; pre-registered sample size; per-game archive	Fourteen games in the Playground, the Track B scorecard, the Arena	That a playground game approximates a real social outcome or that a benchmark score is portable across cities	REAL / INCOMPLETE